

A systematic bibliometric and topic modelling review of crypto financial risk research

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Abstract

The rapid expansion of crypto finance has generated a large and fragmented scholarly literature, making systematic synthesis increasingly necessary. We employ modular BERTopic and bibliometric analysis to examine 891 leading academic and policy publications, integrated with 1,980 security events. The analysis identifies 19 research themes, consolidated inductively into an integrated framework encompassing risk sources, measurement, transmission, hedging, and governance. The findings reveal a thematic shift from early preoccupations with consensus mechanisms and asset valuation toward, particularly since 2018, quantitative analyses of systemic spillovers, DeFi vulnerabilities, and macroprudential governance. This trajectory mirrors the progressive integration of crypto finance with traditional financial systems, as the risk landscape has broadened from protocol-level technical failures to encompass systemic, institutional, and macroeconomic dimensions. It is recommended that future research focus on the following key areas: the integration of granular on-chain data into risk measurement frameworks; the modelling of nested DeFi protocol vulnerabilities and their cascading liquidation dynamics; the hedging functions of high-leverage perpetual futures and prediction markets; and the empirical assessment of the implications of crypto finance for both advanced and emerging economies. We offer fresh perspectives for researchers, practitioners, and policymakers seeking to navigate the multifaceted risks of the crypto ecosystem.

Keywords: Crypto financial risks, BERTopic, Systemic risk, Decentralized finance, Stablecoin

1. Introduction

Crypto finance is a novel financial paradigm using cryptography to verify ownership, facilitate peer-to-peer transfers, and maintain network consensus (Cong and He, 2019; Gomber et al., 2018). It has evolved into a hybrid of decentralized protocols and centralized service providers (The White House, 2025). Its integration with traditional finance exposes operational and regulatory mismatches, creating risks beyond conventional safeguards and challenging monetary sovereignty (Aquilina et al., 2025, 2023). Since 2012, cumulative economic losses from technical risks, such as private key management failures, smart contract vulnerabilities, and hacking incidents, have reached \$37 billion¹. As shown in Fig. A.1 and Table A.1, the system is hit each year by crises from information asymmetries, maturity mismatches, and excessive leverage, as well as shocks from regulatory interventions and macroeconomic liquidity contractions. Crypto's permissionless and anonymous design enables illicit activity while blocking regulatory oversight (Naqbi et al., 2025). Illicit addresses received at least \$40.9 billion in 2024 alone². Figure 1 depicts the amplification of risks through technical coupling, financial leverage, and behavioral biases. This mechanism converts initial risk factors into tangible threats, disrupting finance and economic stability.

[Insert Figure 1 about here]

Research on risks in crypto finance has two principal categories. One category employs a micro-level approach to dissect individual crisis episodes or specific market anomalies (Kyriazis et al., 2020; Li and Zhu, 2026). For instance, regarding crisis events, scholars have investigated the death spiral of the Terra/Luna (Briola et al., 2023;

¹ <https://hacked.slowmist.io/>

² <https://www.chainalysis.com/blog/2025-crypto-crime-report-introduction/>

Liu et al., 2023), the impact of crypto firms' bankruptcies (Conlon et al., 2024a; Martins, 2024), and the USDC de-pegging crisis precipitated by the failure of Silicon Valley Bank (Galati and Capalbo, 2024; Oefele et al., 2024). Regarding market microstructure, scholars have examined market misconduct and anomalous trading behavior, focusing on manipulative tactics such as wash trading (Aloosh and Li, 2024; Amiram et al., 2025), and pump-and-dump schemes (Dhawan and Putniņš, 2023; Li et al., 2025), as well as the influence of social media on crypto prices and the herding effects (Jagolinzer, 2024; King and Koutmos, 2021; Merkley et al., 2024). Research in decentralized finance (DeFi) risks has evolved along two lines. One revolves around protocol and mechanism design, encompassing automated market makers (AMM), liquidity provision, and governance structures (Lehar and Parlour, 2025; Makridis et al., 2023; Park, 2023). The other emphasizes security-focused analyses, which document smart contract exploits, flash loan attacks, and bridge vulnerabilities (Cheraghali et al., 2024; Xu et al., 2025). While these studies have made significant contributions to elucidating specific risk mechanisms, they remain constrained in two respects: in scope, concentration on particular asset classes or isolated crisis episodes; and in temporality, by their reliance on static event snapshots that cannot capture how the risk of crypto finance evolves over time.

The other category operates from a macroprudential, top-down viewpoint. Exemplified by institutions such as the Federal Reserve (the Fed) and the IMF, these studies apply classical financial risk frameworks to conduct examinations of the crypto finance. For instance, Azar et al. (2024) use the Fed's financial stability monitoring framework for digital assets, analyzing risks such as valuation and leverage within the crypto ecosystem through the dimensions of "vulnerabilities" and "shocks". Murcia et

al. (2023) apply a traditional risk taxonomy to analyze market, liquidity, and credit risks, with a focus on emerging markets. Hacibedel (2023) further develops the Crypto-Risk Assessment Matrix (C-RAM) to capture macro-level risk transmission channels. Much of this literature employs a top-down approach, relying on policy-derived deductions or categorical analysis.

Overall, extant research adopts a static perspective, with limited attention to how the risk of crypto finance evolves over time. To address the aforementioned issues, we integrate literature-based topic modelling with the descriptive mapping of historical risk events. First, the modular BERTopic is used to analyze 891 publications from Web of Science and institutional repositories, screened for academic rigor and policy relevance, identifying 19 topics of crypto financial risk. Second, we incorporate 1,980 historical risk events from the SlowMist database, covering attack categories, victim profiles, and economic losses. This allows us to evaluate the alignment between the risk topics prioritized in academic literature and real-world security shocks. Finally, dynamic topic modelling and topic clustering are used to trace temporal shifts in academic priorities over time and the evolution of the field’s methodology, revealing the development trajectory of scholarship on crypto financial risk across the study period.

We contribute to the existing literature in four principal ways. First, to the best of our knowledge, we are the first to apply contextual embeddings and density-based clustering for text mining in crypto financial literature, enabling identification of latent thematic boundaries across 891 papers that keyword co-occurrence methods are unable to resolve. Second, the topic modelling results are consolidated into five aspects spanning risk sources, risk measurement, risk contagion, risk hedging and macro governance. This provides the field with a shared conceptual structure for cross-study

comparison. An integrated analysis of 1,980 security incidents confirms that our proposed risk taxonomy is rational and complete. Third, by applying dynamic topic modelling, this study traces the temporal evolution of research priorities across the study period, revealing a shift in scholarly attention. We reveal a pronounced shift from foundational concerns with consensus mechanisms and asset valuation through quantitative analyses of risk spillovers, price bubbles and DeFi vulnerabilities toward the systemic integration of crypto finance with traditional financial markets, macro-financial stability, and prudential policy design. Fourthly, we review the methodologies relevant to crypto financial risk research. We further propose a future research agenda that emphasizes the integration of granular on-chain data, the tracing of nested protocol logic, and the systematic assessment of crypto finance’s macroeconomic and financial stability impacts on sovereign states.

The paper is organized as follows: Section 2 details the data sources and methodologies. Section 3 analyzes findings across risk sources, measurement, contagion, hedging, and governance. Section 4 traces the evolutionary trajectory of research topics and clustering of methods. Section 5 summarizes the primary findings and proposes future research directions.

2. Methodology

2.1 Data Sources

We construct a comprehensive dataset comprising high-quality academic literature, policy papers, and industrial risk data. We searched the Web of Science database for academic literature up to November 2025. The core search query is formulated as follows: TS=((cryptocurrenc* OR bitcoin OR ethereum OR “digital asset” OR “crypto asset” OR stablecoin OR “algorithmic stablecoin” OR “decentralized finance” OR

“decentralised finance” OR “liquidity pool” OR “smart contract”) AND (risk OR volatility OR crash OR collapse OR contagion OR spillover OR “de-peg” OR hack OR exploit OR “systemic risk” OR regulat* OR “cyber attack” OR bankruptcy OR insolvency)). We limited our sample to journals rated 3 stars or above in the Academic Journal Guide³. Our screening identified 803 journal articles. To encapsulate the perspectives of policymakers, we conducted a comprehensive search of the official repositories of the FED⁴, the Bank for International Settlements (BIS),⁵ and the International Monetary Fund (IMF)⁶ using the keywords “crypto,” “bitcoin,” and “stablecoin” through November 2025. Subsequently, we filtered for research outputs about risk and regulation, yielding 88 pivotal documents. The distribution of the literature is illustrated in Figure 2.

[Insert Figure 2 about here]

We integrate SlowMist Hack Archives data to strengthen the risk framework’s explanatory power. Our dataset comprises 1,980 significant security incidents recorded from 2012 to November 2025. It includes key dimensions such as attack methods, financial losses, and affected ecosystems.

[Insert Figure 3 about here]

Figure 3 depicts Bitcoin prices, USDT supply, blockchain security incidents, and academic research outputs. As the ecosystem’s benchmark asset and core collateral, Bitcoin’s valuation risk, which is measured by metrics such as maximum drawdown (MDD⁷), influences market-wide liquidation risk. Over the years, the escalation in asset

³ <https://charteredabs.org/academic-journal-guide/academic-journal-guide-2024>

⁴ www.federalreserve.gov

⁵ www.bis.org

⁶ www.imf.org

⁷ MDD reflects the maximum price decline from peak to trough over a given period.

valuations and the frequency of risk events exhibit broadly parallel trends. As Bitcoin prices surge, the frequency of blockchain security incidents has shifted from sporadic occurrences to explosive growth, catalyzing an exponential increase in academic output. In 2024, a divergence emerged as prices rose while the frequency of reported risk incidents declined. Publicly disclosed incidents suggest a shift in risk profiles, reflecting ongoing advances in protocol audits and defensive frameworks that may have helped suppress opportunistic attacks during volatile market phases. Nevertheless, losses remained substantial, and academic research on crypto financial risk stayed active.

2.2 Modular BERTopic framework

We employ BERTopic for topic modelling, due to its advantages in addressing characteristics such as dense interdisciplinary terminology and rapid semantic evolution within crypto finance research. Compared to LDA models, which often ignore contextual meaning, modular BERTopic adopts a more nuanced approach. It utilizes the Transformer architecture’s contextual embeddings and density-based clustering algorithms. This enables the identification of latent thematic structures across heterogeneous literature while effectively handling outlier documents (Reisenbichler and Reutterer, 2019; Z. Wang et al., 2024). Consequently, the framework enhances both theoretical clarity and clustering robustness in topic detection. In addition, the model employs a class-based TF-IDF weighting scheme and dynamic modelling techniques

$$MDD = \frac{P_{trough} - P_{peak}}{P_{peak}}$$

In this formulation, P_{peak} denotes the historical peak representing the maximum price attained within a one-year observation window. P_{trough} signifies the trough defined as the minimum price reached following the occurrence of the aforementioned peak.

to trace the evolution of academic research focus over time. Its modular architecture facilitates the integration of Large Language Models (LLMs), enabling the optimization of topic representations. This integrated approach significantly enhances both interpretability and robustness in topic identification (Grootendorst, 2022). The specific research workflow is illustrated in Figure 4.

[Insert Figure 4 about here]

At the embedding stage, the OpenAI text-embedding-3-large model is used to map textual data to a high-dimensional vector space, enabling the precise capture of latent semantic features in the financial domain. At the clustering stage, we leverage uniform manifold approximation and projection (UMAP) for dimensionality reduction, followed by hierarchical density-based spatial clustering of applications with noise (HDBSCAN). This integration facilitates the identification of complex cluster geometries while robustly filtering noise and preserving essential topological structures (McInnes et al., 2020, 2017). The subsequent topic representation employs a class-based term frequency-inverse document frequency (c-TF-IDF) weighting scheme, which extracts salient keywords and mitigates bias caused by class imbalance (Kwon et al., 2024). The parameter configurations for the model are detailed in Table A.2. Subsequent to this, to comprehensively enhance the interpretability and robustness of the identified topics, a rigorous optimization and cross-validation protocol was implemented, comprising the following steps.

First, to enhance the noise-reduction capability of label generation and mitigate potential biases induced by individual outlier documents, we increase the number of representative documents sampled per cluster (`nr_repr_docs`) to ten. This expanded sampling configuration ensures that the LLM processes a broader, more heterogeneous

range of textual samples, thereby generating more comprehensive topic labels and effectively mitigating the risk of model hallucinations.

Second, we conduct systematic comparative experiments to optimize topic representation. Statistical methodologies, specifically maximal marginal relevance (MMR) and part-of-speech (POS) filtering, are compared with frontier LLMs such as GPT, Gemini, and DeepSeek. A comparative analysis reveals that Gemini-generated labels offer the highest clarity and conciseness. Therefore, we utilize Gemini-generated labels for visualization. Alternative representation outcomes are provided in Table A.3.

Finally, to further evaluate the robustness of the clustering structure, we conduct a cross-validation exercise by substituting the text embedding model with gemini-embedding-001. The results of the alternative topic identification are detailed in Table A.4. The close agreement in both the number of topics and their thematic content across the two embedding configurations lends further support to the robustness of the topic identification results in this study. This suggests that the results effectively capture the underlying patterns of crypto financial risk research without being overly sensitive to hyperparameter settings.

3. Risk Analysis Framework

We establish a systematic crypto finance risk framework by integrating findings into an end-to-end risk management process. Based on identified topics, which include 19 core topics (numbered 0–18) and 1 outlier (Topic -1), we utilized cluster analysis on literature embeddings to map these themes across five critical dimensions: risk sources, measurement, contagion, hedging, and governance. Table 1 and Figure 5 summarize these results.

[Insert Table 1 about here]

[Insert Figure 5 about here]

3.1 Risk Sources

Figure 6 maps the key entities and interconnections within crypto finance, providing a basis for identifying the risk sources examined in the following analysis.

[Insert Figure 6 about here]

3.1.1 Taxonomy of Heterogeneous Risk Sources

(1) Endogenous Sources

The four topics identified by BERTopic, specifically Topic 12 (“exchange”, “value”, “bitcoin”), Topic 13 (“blockchain”, “smart”, “supply chain”), Topic 14 (“miners”, “fees”, “transaction”), and Topic 17 (“attacks”, “cyber”, “estate”), focus on endogenous risk sources. They arise from vulnerabilities in cryptographic foundations, consensus mechanisms, and smart contracts, and are coupled with the inherent lack of valuation anchors.

First, private-key-based asset ownership risks permanent loss from poor randomness, storage failures, or operational errors (Erinle et al., 2025; Extance, 2015; Harrast et al., 2022). Quantum computing poses challenges, necessitating evaluations of algorithmic migration pathways (Mascelli and Rodden, 2025). Specifically, Shor’s algorithm (Shor, 1994) compromises the security of elliptic curve cryptography by efficiently solving the elliptic curve discrete logarithm problem (Aggarwal et al., 2018). Hash algorithms exhibit greater defensive resilience against attacks from Grover’s algorithm (Aggarwal et al., 2018; Grover, 1996). Divergent address formats exhibit distinct risk exposures: early P2PK (Pay-to-Public-Key) formats are vulnerable due to

the full transparency of public keys⁸; conversely, P2PKH (Pay-to-Public-Key-Hash) formats hide public keys via double-hashing, revealing them only briefly during the initial transaction, thereby offering greater resilience. The pathways toward Post-Quantum Cryptography (PQC) are not cost-neutral. Instead, they entail substantial trade-offs in computational efficiency and network throughput. Drawing on an assessment (Thaler, 2025), while mainstream post-quantum signature schemes (e.g., Dilithium) afford resilience against Shor’s algorithm, their signature footprints are typically much larger than those of the conventional elliptic curve digital signature algorithm. Transitioning to PQC introduces major externalities, as higher block space demand increases fees and intensive verification delays finality. Addressing vulnerable dormant addresses necessitates a critical trade-off between irreversible asset loss and contentious hard forks to protect or abandon them.

Existing research assesses 51% attack risk by comparing attack costs and payoffs in consensus mechanisms (Budish, 2024; Harvey, 2025). Centralized validation nodes and collusion increase the risk. Proof-of-Work hashrate monopolies or Proof-of-Stake validator alliances can momentarily reverse finality and undermine trust (Albrecher and Goffard, 2022; Eyal and Sirer, 2018; Gans and Halaburda, 2024). Bitcoin’s transition from block rewards to fees faces a risk: low fee levels from on-chain transactions may fail to sustain miner returns, reducing hash rate and lowering attack costs (Bednarek, 2023; Grobys, 2024). Moreover, security heterogeneity exists across blockchain consensus mechanisms. Meanwhile, the fragmented multi-chain landscape hinders

⁸ Early dormant BTC addresses, specifically those exhibiting “Patoshi” patterns, that use the P2PK format and represent the estimated 1.1 million BTC mined by Satoshi Nakamoto (Sergio, 2013), are the most acutely exposed.

interoperability and, via cross-chain activities, exacerbates governance and security risks (Boissay et al., 2022). Table 2 categorizes the endogenous risk sources.

[Insert Table 2 about here]

Second, application-layer risks outlined in Table 3 stem from smart contract flaws and governance privilege abuse (Carapella et al., 2022; Carpentier-Desjardins et al., 2025). The difficulty of static verification for contract code and the unpredictability of dynamic interactions render logic flaws and compiler risks challenging to eradicate (Atzei et al., 2017; Werner et al., 2022). Concurrently, administrative privileges retained by project teams to facilitate upgrades often evolve into centralized single points of failure. The frequent occurrence of theft or abuse highlights the discrepancy between the ideal of decentralization and the reality (Aramonte et al., 2021a; Barbereau et al., 2022).

[Insert Table 3 about here]

DeFi composability amplifies design flaws, giving rise to four categories of risk.. The first is the sustainability of economic models. Many protocols rely on token incentives to attract liquidity (Cong et al., 2022; Corbet et al., 2023). Protocols risk Ponzi spirals needing constant new capital (Cong et al., 2021; Onu et al., 2023). Second, economic exploits originate from the protocol's logical layer. Extant research explores how novel attack vectors, such as flash loans and sandwich attacks, inflict losses on ordinary users (Auer et al., 2022a; Capponi et al., 2025; Lehar and Parlour, 2022). Third, deep protocol nesting propagates vulnerabilities, and failure of a single core protocol (e.g., stablecoin or lending market) can trigger systemic cascading liquidations and liquidity exhaustion (Badev and Watsky, 2023; Lehar and Parlour, 2022). Fourth,

critical infrastructure like oracles and bridges creates ecosystem-wide risks (Cong et al., 2025; Duley et al., 2023).

Finally, most crypto assets lack cash flows. Their value stems from the stream of net transactional benefits (Biais et al., 2023). The valuation of crypto assets follows two paths. Network effect frameworks have gained prominence because evidence confirms a correlation between transaction activity and price dynamics (Cong et al., 2021; Koutmos, 2018a; Sockin and Xiong, 2023). Metrics such as transaction volume and unique address counts validate network utility as a fundamental value driver (Bhambhwani et al., 2023). Production cost models offer an alternative perspective. Literature indicates that fixed sunk costs and energy expenditures support Bitcoin prices from the supply side (Garratt and Van Oordt, 2023; Lambrecht et al., 2024).

While existing literature provides a comprehensive analysis of the endogenous risks, three deficiencies remain unaddressed. First, the dynamic game-theoretic interactions between developers and communities during protocol upgrades and rule modifications have not been extensively discussed. Second, there is a paucity of research on how significant changes to protocols or public chain mechanisms affect the valuation logic of crypto assets. Third, the composite vulnerabilities arising from the coupling of diverse DeFi protocols represent a significant research gap.

(2) Institutional Sources

The three topics identified by BERTopic, specifically Topic 9 (“icos”, “initial coin”, “offerings”), Topic -1 (“price”, “bitcoin”, “exchanges”), and Topic 1 (“stablecoins”, “stability”, “financial”), focus on institutional risks, such as centralized exchanges (CEXs), stablecoin issuers, and crypto investment firms. These risks arise from core functions: custody, clearing, and credit intermediation. In the absence of

stringent regulatory oversight and lender-of-last-resort (LOLR) (Samuel et al., 2025), these entities have functionally morphed into “shadow banks” (Olk and Miebs, 2025).

CEXs operate as vertically integrated conglomerates, consolidating custody, matching, clearing, and proprietary trading. This creates conflicts of interest where platforms exploit informational advantages for insider trading or asset misappropriation, fostering principal-agent problems (Azar et al., 2024; Vidal-Tomás, 2025). Empirical evidence suggests wash trading is endemic in certain CEXs, with upwards of 70% of reported volume artificially generated (Cong et al., 2023b). Limited regulatory oversight allows predatory strategies like pump-and-dump schemes to flourish unchecked, transferring wealth from retail investors to manipulators (Ardia and Bluteau, 2024; Hamrick et al., 2021). Moreover, CEX liquidation mechanisms are self-reinforcing. Without circuit breakers or capital buffers, adverse price shocks trigger cascading liquidations, evaporating liquidity during tail-risk events and distorting prices across markets (Alexander et al., 2023; Cornelli et al., 2023). The full risk components are summarized in Table 4.

[Insert Table 4 about here]

Stablecoin issuers operate with business models analogous to those of money market funds (Melo et al., 2022), implying that theories of maturity mismatches and liquidity runs in shadow banking intermediaries are applicable to these entities (Gorton and Zhang, 2021; Pozsar, 2014). Table 5 categorizes the risks of stablecoin issuers.

[Insert Table 5 about here]

Asset-side risks stem from issuer agency conflicts, notably a search-for-yield propensity. Issuers may exploit ambiguous disclosure rules to replace assets with riskier instruments, like loans or commercial paper, to capture interest rate spreads. This

fosters moral hazard (Maex and Slavov, 2025). The literature further underscores the classification of stablecoins as information-sensitive assets and posits that the pegging mechanism is susceptible to rapid disintegration should the fair value of reserve assets fall below a critical threshold (Ahmed et al., 2024; Aldasoro et al., 2023).

On the liability side, literature emphasizes the reliance of stablecoin demand upon leveraged speculative activities (Gorton et al., 2025). The velocity of maturity transformation and the architectural design of redemption channels represent critical determinants of run risk. Ma, Zeng and Zhang (2023) dissect the impact of redemption fees and arbitrage efficiency and find that while enhanced arbitrage efficiency strengthens stablecoin price stability, it simultaneously exacerbates run risk, suggesting an inherent trade-off between minimizing run risk and maintaining price stability. The “Co-instability” framework illustrates how run risk spreads via shared liquidity pools and arbitrage paths, allowing localized protocol failures to escalate into systemic dangers that jeopardize the entire stablecoin framework (Anadu et al., 2023; Gregory et al., 2024; Oefele et al., 2024).

Crypto investment institutions encompassing hedge funds, venture capital, market makers, and centralized crypto lending platforms not only facilitate liquidity in crypto markets but also maintain intricate credit-debt linkages characterized by direct balance-sheet interconnectedness (Bardoscia et al., 2021). The risks are delineated in Table 6. The crypto issuance mechanisms have evolved from public-facing ICOs toward an institution-dominated “low float/high fully diluted valuation” model. Early models distributed risk directly and diffusely across retail participants. In contrast, new models employ engineered tokenomics, phased unlocking schedules, and preset institutional valuations to achieve delayed, structured risk transfer (Dhawan and Putniņš, 2023;

Drobetz et al., 2024). This engenders inequities in token allocation and facilitates governance capture (Aramonte et al., 2021a; Gan et al., 2023).

Centralized crypto lenders fund long-term crypto assets or inefficient DeFi protocols with short-term redeemable liabilities. Patel and Rose (2023) identify a “liquidity-solvency” death spiral without safety nets: when asset prices breach collateral thresholds, panic redemptions rapidly drain liquidity, leading to immediate insolvency and cascading repercussions.

[Insert Table 6 about here]

Literature has identified institutional risks, but without standardized disclosure, granular data on internal leverage and related-party transactions remain unavailable. Amid fragmented global regulation, these institutions have used vertical integration to consolidate market dominance and expand assets rapidly. Yet forward-looking research on their systemic impact on traditional finance and broader capital markets remains scarce.

(3) Behavioral Sources

The two topics identified by BERTopic, specifically Topic 8 (“perceived”, “intention”, “consumers”) and Topic 6 (“sentiment”, “investor”, “news”), focus on the roles of behavioral biases, participant risk perception, and shifts in collective sentiment in the formation and contagion of risks.

Topic 8 research highlights the retail-centric nature of crypto markets, where retail investors exhibit extreme turnover and high speculative sensitivity (Kogan et al., 2024). The decision-making logic is imbued with a strong “gambling” ethos (Chava et al., 2025) and manifests a pronounced preference for “lottery-like” payoffs (Han, 2026).

By pursuing low-probability yet high-magnitude returns, investors catalyze severe price deviations from intrinsic valuations (Grobys and Junttila, 2021).

Topic 6 focuses on “investor attention” in risk pricing under social media dynamics. Metrics like Google Trends and social media activity act as indicators that precede price volatility (Said et al., 2023; Tash et al., 2024). Research indicates that crypto influencers leverage social media to significantly modulate investor risk-taking thresholds (Merkley et al., 2024). Attention triggers indiscriminate herding among retail investors (Arnold et al., 2022). Such collective behavior, rooted in social identity cohorts, not only presents formidable regulatory challenges (Jagolinzer, 2024) but also amplifies procyclical market volatility via high-leverage mechanisms during market stress. Consequently, academic discussion regarding the imposition of retail leverage constraints warrants attention (Heimer and Simsek, 2019).

Although studies identify sentiment spillovers and behavioral biases in crypto markets, they often treat social media only as a proxy for sentiment, neglecting how social network consensus reshapes investor logic. Future research could utilize LLMs for detailed semantic analysis and combine information diffusion models with on-chain data.

(4) Exogenous Sources

The two topics identified by BERTopic, specifically Topic 0 (“energy”, “green”, “connectedness”) and Topic 12 (“exchange”, “value”, “bitcoin”), focus on external risks, as detailed in Table 7.

[Insert Table 7 about here]

Physically, crypto finance relies on centralized cloud services and power infrastructure (Ethereum Foundation, 2025; Guidi et al., 2025; Sun et al., 2022). Cloud

service disruptions often lead to on-chain confirmation delays and a deterioration in network security. Energy policies directly affect hash power. Fluctuations in energy demand and prices exert spillover effects on crypto assets (Okorie, 2021; Okorie and Lin, 2020; Pham et al., 2022).

Regarding policy, first, regulatory actions (e.g., AML reviews or security attribute designations) increase market volatility and trigger sharp price revaluations (Auer and Claessens, 2020; Griffith and Clancey-Shang, 2023), giving rise to regulatory spillover effects that lead to an increased prevalence of low-rated ICOs in other jurisdictions (Bellavitis et al., 2022). Second, monetary policy tightening, such as the Fed's 2022 rate hikes, pressures highly leveraged crypto institutions by contracting liquidity, triggering market adjustments (Che, 2023). Simultaneously, the crypto market demonstrates rapid and intense reactions to geopolitical signals and major shocks. Crypto assets spill risk to external markets under normal conditions but receive reverse spillovers from traditional markets during financial stress (Iyer, 2023).

While research has explored these external factors, there is a lack of quantitative assessments and mechanistic research on the impact of various centralized cloud services on the security of crypto finance. Additionally, the resource competition between AI development and Bitcoin mining is a critical area of concern, encompassing power infrastructure, semiconductor supply, and specialized human capital.

3.1.2 Risk Mapping and Risk Frontier Evolution

Using data from the SlowMist database spanning 2012 to 2025, we map the aforementioned risk sources. Figure 7 illustrates the attack methods associated with risk sources. Although these attacks stem from technical and operational vulnerabilities, they can escalate and evolve into other risk categories. Early-stage losses were

predominantly due to security vulnerabilities in the consensus mechanisms, such as hashrate attacks targeting PoW. Smart contract vulnerabilities recur at high frequency, providing corroboration for the endogenous risk sources identified in this study. This suggests that logical fragility in code remains the most fundamental endogenous threat to the system. At the same time, the escalating scale of losses from phishing and social engineering attacks is consistent with the institutional risk sources proposed. The major losses recorded to date have originated from project teams and CEXs. This indicates that the primary source of risk is shifting from code-level flaws to human and procedural vulnerabilities. It highlights the vulnerability of private keys as a single point of failure. Further, the prevalence of rug pulls underscores critical moral hazard and agency costs, compounded by regulatory voids and opaque disclosure. Notably, this is difficult to resolve through technical means alone.

[Insert Figure 7 about here]⁹

Data further reveal spatiotemporal dynamics in risk sources, as shown in Figure 8. Risk hotspots shift across sectors as capital concentrations change. Attack targets have expanded from isolated risk points within CEXs during the early stages to active decentralized ecosystems such as Ethereum and BNB Smart Chain. While the absolute frequency of attacks on cross-chain bridges is not the highest, their structural role as multi-chain asset nexuses results in extreme individual loss magnitudes, exhibiting pronounced tail-risk characteristics. The “Other” category highlights an expanding range of targets, including emerging stablecoins, decentralized AI networks, and

⁹ The bubble size represents the annual number of hacking incidents. The numerical value inside the bubble indicates the total amount stolen each year. This dataset is based on token prices at the time of each incident. Due to factors such as token price volatility, undisclosed incidents, and the exclusion of losses incurred by ordinary users, the actual losses are likely higher than the statistical results indicate.

infrastructure such as oracles and solvers. It also records targeted attacks on high-net-worth individuals and projects' social accounts.

[Insert Figure 8 about here]

3.2 Risk Measurement

The three topics identified by BERTopic, specifically Topic 5 (“options”, “volatility”, “models”), Topic 10 (“flows”, “capital”, “crossborder”), and Topic 16 (“explosive”, “test”, “simulations”), focus on risk measurement methodologies. Our review of the literature shows that the risk measurement framework mainly includes three dimensions: extreme market risk metrics, cross-border capital flow measurements, and asset bubble diagnostic tests.

For endogenous technical and institutional risks, extreme market risk metrics quantify systemic price collapses from protocol vulnerabilities, excessive leverage, and exogenous shocks. Distinct stylized facts in market data manifest as non-normal return distributions, where high leverage amplifies localized shocks into extreme volatility and heavy-tailed price collapses. Persistent market uncertainty drives volatility clustering and long-range dependence, exacerbating systemic fragility across interconnected markets (Borri and Shakhnov, 2023; Näf et al., 2019). Literature has adopted integrated frameworks, ranging from hybrid GARCH-family models to combinations of GARCH with machine learning. For instance, Vine Copula-GARCH frameworks and hybrid GARCH-LSTM models have been applied to decompose multi-dimensional tail dependencies and the non-linear persistence of idiosyncratic shocks inherent in crypto market dynamics (Bedoui et al., 2023; B. Han et al., 2025). Since crypto markets often crash, risk frameworks based on extreme value theory offer more reliable tail risk estimates. Many studies apply the generalized Pareto distribution to

model the tails of return distributions, combining it with the conditional VaR to quantify potential losses transmitted from crypto assets to broader financial markets (Borri, 2019; Rehman and Wu, 2025; Umar et al., 2024). CAViaR and Expected Shortfall (ES) are utilized to capture dynamic quantile evolution and expected tail losses (Kwon, 2020; Patra and Gupta, 2023). Furthermore, refined metrics such as Vulnerability-CoVaR, MVaR, and Financial Risk Meter (FRM) have been introduced to quantify systemic fragility and exchange resilience (Vidal-Tomás, 2025; Waltz et al., 2022; R. Wang et al., 2024). Various user-level metrics and pool-level resilience metrics are used to assess liquidation risk in DeFi (Heimbach and Huang, 2024). For stablecoins, devaluation risk is estimated via the futures-spot basis, while transaction velocity signals potential run behavior (Eichengreen et al., 2025).

Since cross-border crypto flows originate from technical risk sources (permissionless blockchains) and institutional risk sources (CEXs and issuers), traditional Balance of Payments statistics cannot adequately capture them (Cardozo, 2024). The literature adopts two approaches to address this measurement gap. One research pathway utilizes off-chain and peer-to-peer data to identify “crypto vehicle trades”, which means rapid conversions between local currency, crypto assets, and foreign currency. This method quantifies how crypto assets facilitate capital flight to bypass capital controls (Cerutti et al., 2025; Graf Von Luckner, 2024). The other pathway integrates on-chain data with proxy variables to capture capital flows. This approach identifies entities by clustering blockchain addresses. It then infers the geographic distribution of user populations by incorporating web traffic or application usage data. Finally, these elements are combined to estimate the structure of global capital flow networks (Auer et al., 2025). Furthermore, by analyzing the domain

characteristics and interaction patterns of self-custodied wallets, research can more precisely trace the cross-border flows of international stablecoins (Reuter, 2025). Stablecoins now carry nearly half of cross-border crypto capital flows, with mobility far less constrained by distance than in traditional finance and distinct sensitivity to global financing conditions (Auer et al., 2025).

Reflecting the behavioral risk sources identified, bubble-detection methods quantify valuation risks from irrational exuberance, where sentiment, narratives, and speculation decouple prices from fundamentals (Cai et al., 2021; Whitehouse et al., 2025). The literature employs two approaches. The first relies on recursive regression-based right-tailed unit root tests, such as the GSADF test within the PSY framework. The second uses quantile regression to analyze the structural characteristics of price distributions (Chen et al., 2024; Harvey et al., 2020). Research indicates statistical asymmetries in bubble dynamics. The expansion phase features explosive growth as the autoregressive coefficient (the “root”) exceeds unity, whereas the collapse phase shows pronounced mean reversion with the coefficient dropping below one (Pavlidis, 2025). The shift in statistical regime signals an abrupt transition in market consensus, from over-expansion to liquidity exhaustion. Further analysis reveals significant “co-bubble effects” across crypto assets (Adewuyi et al., 2024; Chen et al., 2024). Recent research has introduced advanced geometric computational methods, such as generalized visible curvature, to enhance the sensitivity of statistical frameworks in detecting extreme volatility regimes (Zhang et al., 2024).

Despite these advances, several limitations remain. First, existing frameworks assess risk sources in isolation, failing to adequately identify the interconnectedness between market volatility, cross-border capital flows, and asset bubbles, including how

large capital movements catalyze bubbles. Second, existing research has made limited progress in integrating multi-source attribute data and granular on-chain entity labels from analytics firms. Furthermore, more attention is required for precise assessment of liquidity depth in exchanges. Additionally, the measurement of leverage levels remains underexplored, and the nonlinear amplification effects of high-leverage exposures during price jumps warrant further investigation.

3.3 Risk Contagion

The four topics identified by BERTopic, specifically Topic 1 (“stablecoins”, “stability”, “financial”), Topic 3 (“volatility”, “connectedness”, “spillovers”), Topic 7 (“futures”, “spot”, “arbitrage”), and Topic 11 (“defi”, “lending”, “finance”), focus on the analysis of risk transmission channels. Based on identified literature, risk transmission pathways are categorized into contagion within the crypto market and contagion to traditional finance. The specifics are detailed in Table A.5.

Risk contagion within crypto finance is driven by three mechanisms: correlated asset price movements, market structural frictions, and technology-embedded dependencies. First, crypto assets exhibit a profound dependence on a unified risk appetite and liquidity environment. Most crypto assets track Bitcoin’s price and market sentiment. Common factors make crypto asset returns highly correlated (Allen et al., 2022; Jiang et al., 2023; Liu et al., 2022). Scholars utilize the Diebold-Yilmaz spillover index and TVP-VAR models to confirm significant, time-varying volatility spillovers across crypto assets. Such spillovers intensify under extreme market pressure and exhibit asymmetric panic contagion characteristics (Aslanidis et al., 2021; Corbet et al., 2020; Koutmos, 2018b). Second, the crypto market is highly fragmented. Latency and technical barriers impede cross-platform arbitrage, allowing price disparities to persist

across exchanges (Hautsch et al., 2024; Schmeling et al., 2023). Under liquidity shocks, leveraged arbitrageurs transmit liquidation pressure to unaffected venues, amplifying systemic vulnerability (De Blasis and Webb, 2022; Makarov and Schoar, 2020). Third, DeFi protocols may create algorithmic contagion channels. Through re-staking and re-collateralization, assets establish deep cross-protocol dependencies. This allows underlying price volatility to spark automated liquidation spirals, diffusing stress across the ecosystem (Yin et al., 2025).

Crypto risks transmit to traditional finance via institutional crypto exposure, stablecoin reserve assets, and stablecoin-driven disintermediation. First, risk transmission manifests as balance sheet erosion for entities holding crypto assets and impacts investors through wealth effects (Auer R. et al., 2022b). Second, stablecoin issuers use highly liquid assets, such as Treasury bills, repurchase agreements, and bank deposits, as their primary reserves. Fire-sale activities in stressed environments can disrupt liquidity and influence interest rates in short-term financing markets (Ahmed and Aldasoro, 2025; Yadav and Malone, 2025), potentially leading to run risks for banks. Third, research suggests that stablecoin expansion facilitates the migration of bank deposits into stablecoin reserves (Malloy and Lowe, 2021). This compresses bank net interest margins, raising the probability of banking crises (Chen and Phelan, 2025). It also weakens the credit provision capabilities of banks (Liao and Caramichael, 2022; MacDonald and Zhao, 2022), and consequently interferes with the transmission of monetary policy (Clouse, 2024). Furthermore, research constructs a two-country New Keynesian model to demonstrate that in small open economies where stablecoins are widely adopted, adverse shocks exacerbate domestic currency substitution and capital outflows (Copestake, 2023).

Several issues remain underexplored. First, how tokenized deposits alter traditional bank run dynamics remains unclear. Second, the different impact of stablecoins on monetary policy transmission between small and large economies remains under-researched. Third, currency substitution risks and the erosion of sovereign monetary policy efficacy are validated in theory and simulations but lack empirical evidence. Future research should prioritize crypto finance's impact on real-economy credit supply, employment, and output.

3.4 Risk Hedging

The three topics identified by BERTopic, specifically Topic 2 (“returns”, “learning”, “momentum”), Topic 4 (“gold”, “safe”, “safehaven”), and Topic 18 (“diversification”, “portfolios”, “optimization”), focus on risk hedging, aiming to mitigate extreme price volatility and tail risks through diverse financial instruments and quantitative strategies.

Research on hedging instruments has focused on their effectiveness. First, precious metals, particularly gold, are identified as stable hedging instruments against policy and price uncertainty within crypto markets. Analysis employing DCC-GARCH models reveals a persistent positive correlation between gold and the crypto asset uncertainty index (UCRY), which significantly mitigates systemic portfolio exposures (Hassan et al., 2021). Second, stablecoins function as safe havens, as investors frequently rotate into them during extreme Bitcoin price drops (Baur and Hoang, 2021). Furthermore, Bitcoin and Ethereum have been demonstrated to serve as instruments for hedging economic policy uncertainty (EPU) in the short term (He et al., 2024; Umar et al., 2021). These assets also serve as hedging vehicles against the devaluation of major fiat currencies in intraday trading (Urquhart and Zhang, 2019). Beyond the

aforementioned products, research also extends to using crypto assets to hedge real estate investment trust (REIT) portfolios (Odusami and Akinsomi, 2024) and to address risk exposures in energy commodities such as crude oil (Selmi et al., 2018; Syuhada et al., 2022). A broader body of research underscores that the hedging capacity of crypto assets is highly contingent on shifts in macroeconomic and market sentiment cycles (Conlon et al., 2024b; Li and Miu, 2023; Wen et al., 2022).

To enhance hedging, research prioritizes strategies using hybrid models that integrate deep learning with advanced econometrics. Regarding return and volatility forecasting, researchers integrate multi-source data, including market trend factors, on-chain activities, social media sentiment, and code repository updates (Fang et al., 2024; Fieberg et al., 2025; Gurgul et al., 2025; Lucchini et al., 2020). Reinforcement learning algorithms, such as the Multi-level Deep Q-Network, enable dynamic optimization of trading decisions and improve Sharpe ratios (Otabek and Choi, 2024). TVP-FAVAR networks and Expectile VaR (EVAR) enable real-time hedge ratio optimization and tail risk control, transcending the mean-variance paradigm to mitigate market contagion and downside losses (Ali et al., 2024; Antonakakis et al., 2019; Patel et al., 2024; Trichilli et al., 2025).

Second, liquidity constraints affect hedging efficacy. Zhang and Li (2023) demonstrate that liquidity risk commands a significant negative premium in the cryptocurrency cross-section. Consequently, optimization models should account for transaction costs to avoid slippage losses during rebalancing. The Liquidity-Bound Risk-Return Optimization (LIBRO) model, introduced by Trimborn, Li and Härdle (2020), operationalizes this by incorporating explicit liquidity thresholds into portfolio construction. The practical implementation of such models, however, depends on

accurate liquidity measurement. Brauneis, Mestel, Riordan and Theissen (2021) show that the choice of liquidity proxy is context-dependent. While the Amihud measure is useful for cross-sectional analysis, high-low spread estimators are superior for capturing time-series variation in transaction costs. Therefore, advancing hedging strategies requires both liquidity-constrained optimization frameworks and empirically supported liquidity metrics.

Several areas in crypto hedging deserve more attention. Perpetual futures hedging under extreme conditions lacks systematic investigation. The integration of discrete prediction market instruments such as binary options into continuous portfolios is nearly unexplored. Promising future directions include using LLMs to interpret unstructured risk signals and developing autonomous agents for portfolio hedging.

3.5 Regulation and Governance

The two topics identified by BERTopic, specifically Topic 10 (“flows”, “capital”, “crossborder”), and Topic 15 (“governance”, “blockchain”, “accounting”), focus on regulation and governance of crypto finance.

First, global regulatory fragmentation fosters arbitrage and capital flight while hindering tax enforcement. Crypto firms exploit “jurisdictional jumping” to bypass domestic oversight, sparking a global regulatory “race-to-the-bottom” (Aramonte et al., 2021b; Auer and Claessens, 2018). Crypto assets’ peer-to-peer structure facilitates capital control circumvention. In unstable economies, these unmediated flows bypass traditional gatekeepers, destabilizing national balances of payments (Copestake, 2023; Kokenyne et al., 2022). Anonymity and disintermediation complicate tax administration and oversight, hindering transaction tracking and eroding the tax base. Market volatility and a lack of standardized valuation methodologies prevent reliable

fair value determination, obstructing precise tax collection (Baer, 2023; Cong et al., 2023a; Makarov and Schoar, 2022; Waerzeggers, 2023).

Second, DeFi lacks central intermediaries for “know your customer”, creating regulatory hurdles and complicating accountability for algorithmic code (Carapella et al., 2022). Encoding regulatory rules into protocols enables embedded supervision, automating compliance and shifting from ex post enforcement to ex ante prevention (Auer, 2019).

Third, crypto finance displays issuance and governance inequalities. Market hierarchy and heterogeneous participant strategies amplify existing disparities. Bindseil and Schaaf (2024) develop a dynamic model based on an overlapping generations framework to simulate wealth redistribution between early adopters and late entrants. As a non-productive asset, Bitcoin’s zero-sum nature facilitates an irreversible consumption transfer from latecomers to early adopters, potentially entrenching social stratification. In decentralized autonomous organizations (DAOs), token-weighted voting facilitates monopolization by whales (Sun et al., 2024). This creates de facto oligarchies masked by nominal voting. Governance attack risks rise, including treasury misappropriation and malicious parameter recalibration (J. Han et al., 2025; Sapkota, 2025). Cong et al. (2023c) challenge the purported democratization of DeFi by analyzing on-chain metrics (transaction fees relative to value, failure rates, latency) and concentration metrics (Gini coefficient, Herfindahl-Hirschman Index, Shannon Entropy) to quantify inequality in Ethereum. Institutional and whale advantages in resources, information, and technology exacerbate adverse risk selection by preempting exit channels, while retail investors bear systemic risk (Liu et al., 2023), exposing distributional tensions under the guise of technological neutrality. Finally, a small but

growing body of crypto-accounting literature enhances risk governance by improving the transparency and measurability of crypto-related exposures (Anderson et al., 2025; Luo and Yu, 2024).

While literature focuses on technical and legal analysis, it ignores crypto regulation as sovereign rivalry: a statecraft weapon for monetary dominance in a fragmented order. Future research should examine how regulatory divergence across jurisdictions shapes global monetary power dynamics and the potential for currency bloc formation.

4. Research Trajectory

4.1 Analysis of Topic Evolution

Figure 9 presents the results of the dynamic topic analysis. During 2014-2017, sporadic studies addressed topics such as mining economics (Topic 14) and Bitcoin valuation analysis (Topic 12). This period was characterized by heightened exchange credit risks (following the 2014 Mt. Gox bankruptcy), rapid advances in mining hardware, and a notable rise in Bitcoin's hashrate.

[Insert Figure 9 about here]

Within a year of Ethereum's 2015 launch, the 2016 "The DAO" hack first revealed to academia that smart contract code vulnerabilities could cause systemic losses. Risk research focused on examining consensus mechanisms risk, rather than exploring financial contagion. Additionally, the presence of multiple CEXs motivated investigations into arbitrage and market efficiency (Topic 7). This phase centered on the identification of endogenous risk sources.

From 2018 to 2020, following the launch of Bitcoin futures on the CME, research focus shifted toward quantitative frameworks for measuring risk and spillover effects

in crypto finance. Studies increased on futures-spot arbitrage (Topic 7) and volatility spillover (Topic 3). The boom and subsequent bust of ICOs spurred research on investor sentiment (Topic 6) and ICO-related topics (Topic 9). During the bear market, interest grew in Bitcoin's safe-haven status relative to gold (Topic 4), tail risk modelling (Topic 5), and bubble detection (Topic 16). Following the COVID-19 pandemic and the implementation of large-scale economic stimulus policies, research examining the role of crypto assets in portfolio management (Topic 18) gradually increased.

In 2021, as the bull market began, prediction methodologies (Topic 2) emerged as a stable and prominent research focus. Subsequent investigations shifted toward analyzing market microstructure anomalies, risk formation, and contagion mechanisms. Following "DeFi Summer," research pivoted to examine risks in DeFi protocols (Topic 11) and the impact of cyberattacks on crypto assets (Topic 17). Simultaneously, China's 2021 ban on mining and Ethereum's 2022 transition to PoS were followed by a decline in mining-related studies (Topic 14). Post-2021, research on cryptocurrency and energy market linkages surged (Topic 0), mirroring heightened academic focus on environmental themes during the global low-carbon transition. Viral social media narratives of rapid wealth and the Dogecoin-led meme coin frenzy intensified research into investor behavior and attention (Topic 8). Finally, Terra/Luna collapse, FTX bankruptcy, and USDC de-pegging preceded a surge in stablecoin risks and financial stability studies (Topic 1).

By 2024-2025, the approval of Bitcoin ETFs and the implementation of regulations like MiCA and the U.S. GENIUS Act accelerated the convergence between crypto and traditional finance. This shift pivoted research toward macroprudential assessment, with a focus on stablecoins, financial stability (Topic 1), and cross-border

capital flows (Topic 10). Academic and regulatory efforts prioritized addressing systemic risks from crypto-banking integration and developing robust AML and cross-border frameworks.

4.2 Research Methodology Analysis

Figure 10 illustrates the topic-based clustering of prevalent methodologies. GARCH-family models remain widely adopted for basic volatility measurement (Topics 4, 5). The TVP-VAR model has become a mainstream tool, extensively applied to quantify dynamic spillover effects (Topic 0, 3). This approach effectively addresses structural parameter shifts during extreme market conditions. Scholars use Granger causality tests (and nonlinear variants) to establish causal relationships between crypto assets and safe-haven assets like gold (Topic 4) and to link them to macro-financial stability (Topic 1). Given rising regulatory scrutiny, event study methodology is frequently used to isolate the short-term market impact of regulatory news (Topic 10) and cyberattacks (Topic 17) from market noise.

To accurately characterize tail risk contagion, researchers have introduced Quantile Regression and the CAViaR model (Topics 0, 7). These methods shift the focus from conditional mean dependencies to the tails of the distribution, thereby allowing for the precise measurement of how risks amplify during market crashes. To identify bubbles and distinguish normal fluctuations from speculative explosions, research frequently utilizes right-tailed unit root tests (GSADF/BSADF) alongside their explosive root detection procedures (Topic 16). This approach uses recursive windows to precisely locate bubble inception and burst points, becoming the standard paradigm in this field.

[Insert Figure 10 about here]

Furthermore, in studies on cross-border capital flows and illicit finance regulation (Topic 10), fixed-effects panel regressions and gravity models are used to control for country-level heterogeneity, enabling the evaluation of regulatory policy effectiveness at the macro level.

Concerning the unique risks in DeFi protocols and blockchain, traditional methods are often inadequate. Some research integrates on-chain transaction analysis with game-theoretic modelling (Topics 11, 13, 14). Mixed-methods approaches have emerged for sociological issues like investor behavior and DAO governance. Qualitative methods like PLS-SEM, fsQCA, and Q-methodology (Topics 8, 12, 15) address quantitative models' limitations in explaining investor mental accounting and governance, revealing crypto finance's socio-technical attributes.

5. Conclusion

A comprehensive analysis of 891 publications on crypto financial risks is conducted using bibliometric and topic modelling methods. A total of 19 research topics are identified and integrated into a framework encompassing risk sources, measurement, transmission, hedging, and governance. The dynamic topic modelling indicates a significant three-phase shift in scholarly attention. Prior to 2018, research concentrated on foundational issues, especially consensus mechanism design and crypto asset valuation. From 2018 onwards, the field transitioned towards quantitative analyses of risk spillovers, price bubbles, and market contagion, accompanied by an escalating focus on DeFi vulnerabilities and crypto-energy interdependencies. The most recent phase has been characterized by a shift towards the integration of crypto finance with traditional finance, macro-financial stability, and prudential policy design.

Crypto financial risk research leaves several critical areas for further inquiry. Within risk sources and transmission, the interplay between technical vulnerabilities, behavioral biases, and exogenous shocks remains less documented. Future priorities include modeling DeFi composability through smart contract logic to quantify stress propagation across protocols. For risk measurement, the field benefits from moving beyond secondary market metrics toward the systematic utilization of granular on-chain records and entity labels to track capital flows. Promising avenues in risk hedging focus on how high leverage in perpetual futures amplifies price collapses during market jumps and evaluate the integration of prediction market instruments into portfolios. Finally, macroprudential governance research extends to the systemic impact of stablecoins on banking stability and monetary policy transmission, particularly for small economies facing risks of currency substitution and eroded financial sovereignty.

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Appendix

[Insert Fig. A.1 about here]

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[Insert Table A.2 about here]

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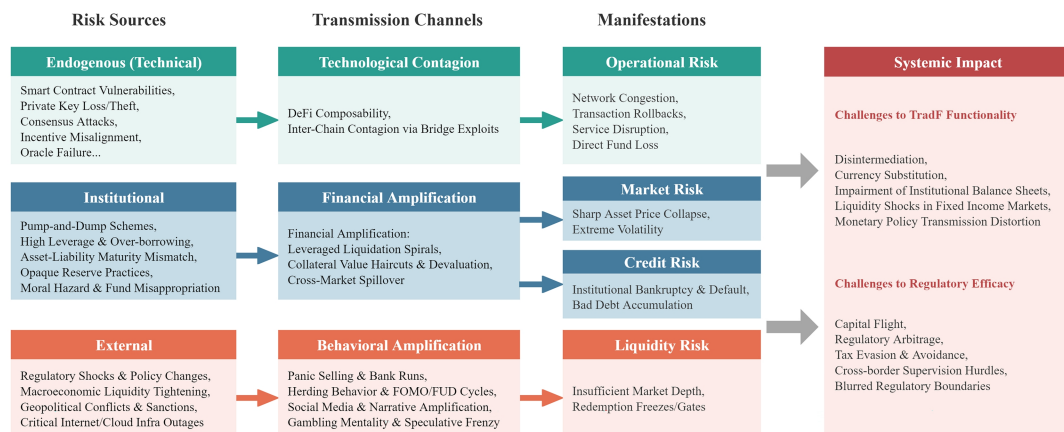


Figure 1. Contagion pathways and amplification mechanisms of crypto financial risks

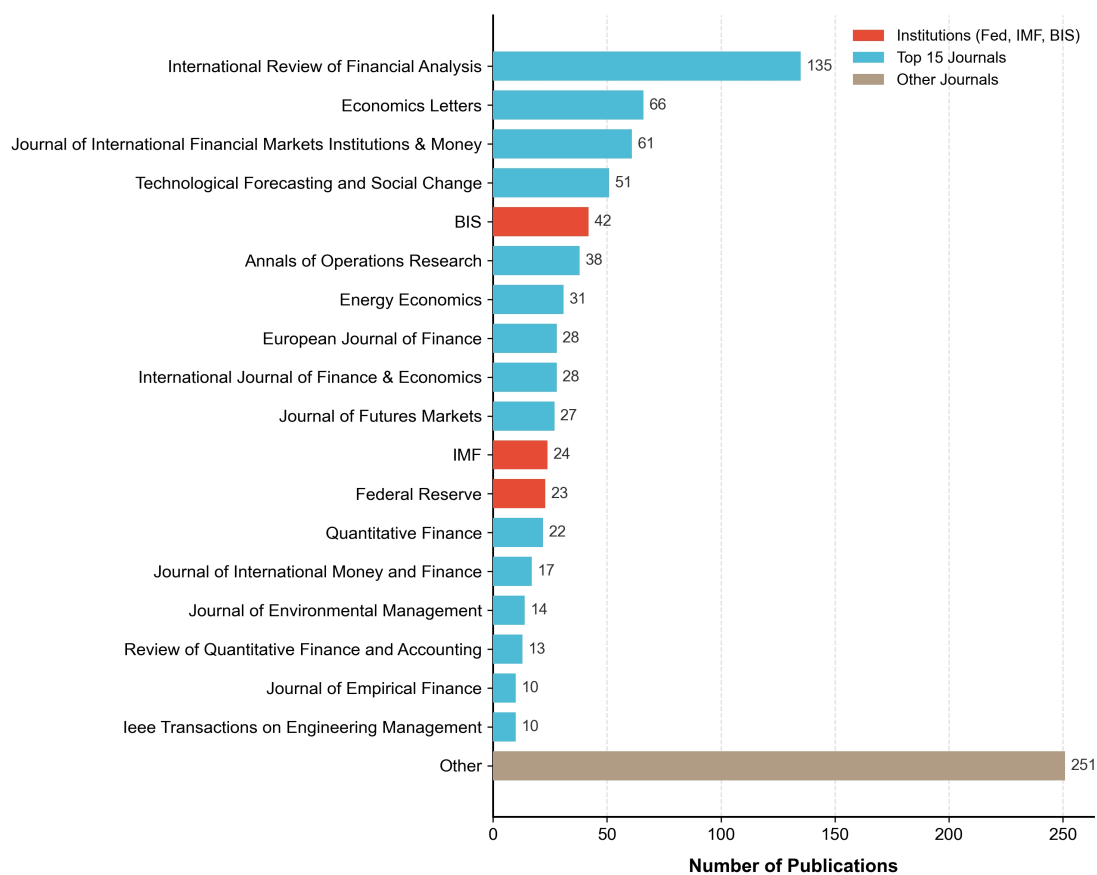


Figure 2. Journal and institutional distribution of the literature

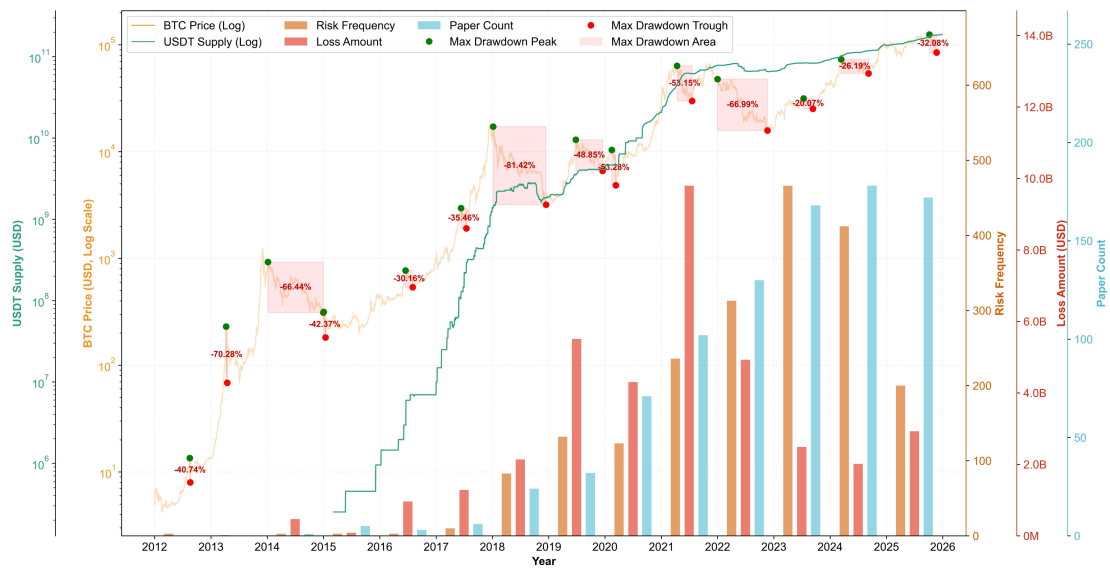


Figure 3. Crypto financial risk and academic research outputs

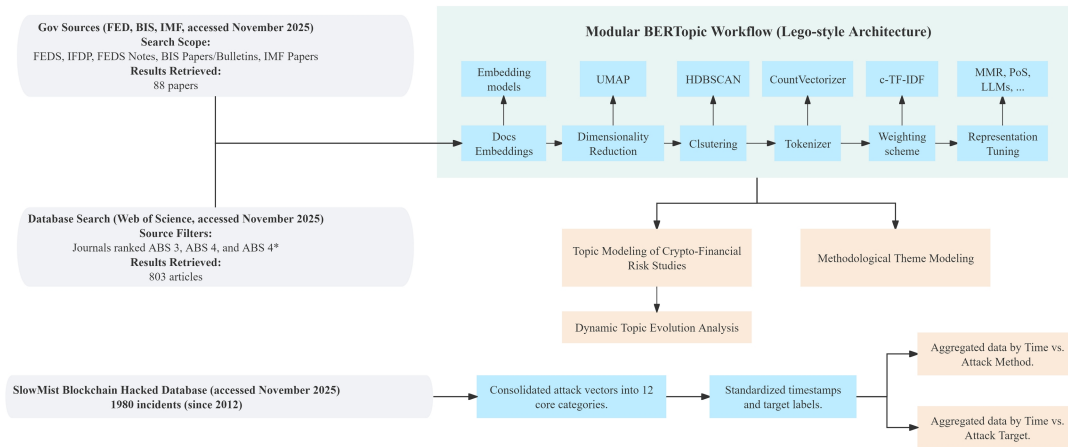


Figure 4. Methodological workflow for topic identification in crypto financial risk management

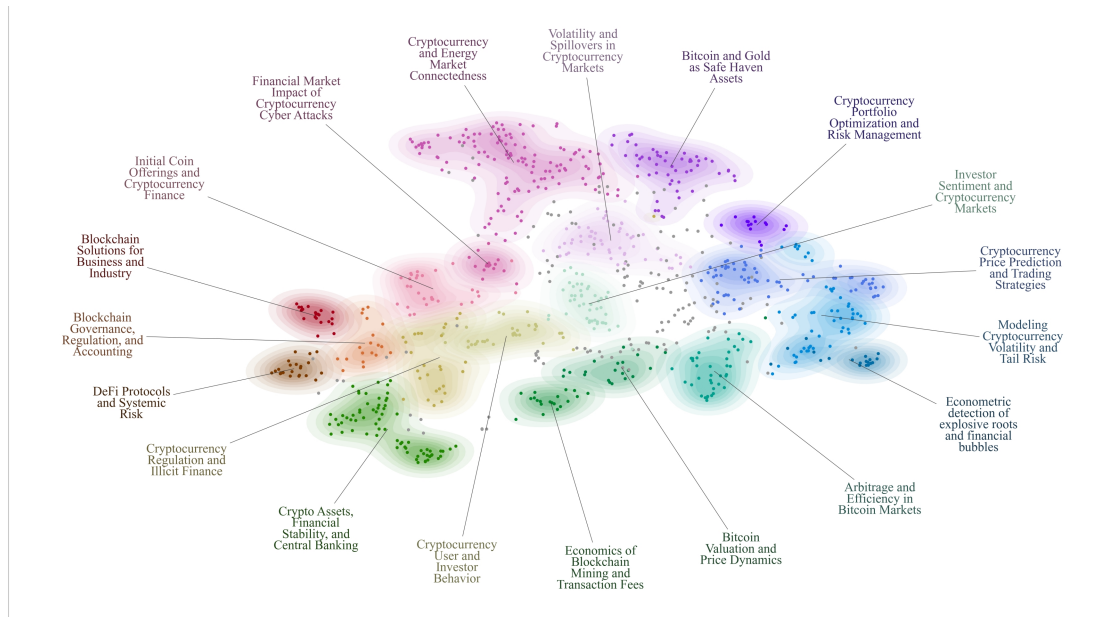


Figure 5. Cluster distribution of crypto financial risk topics

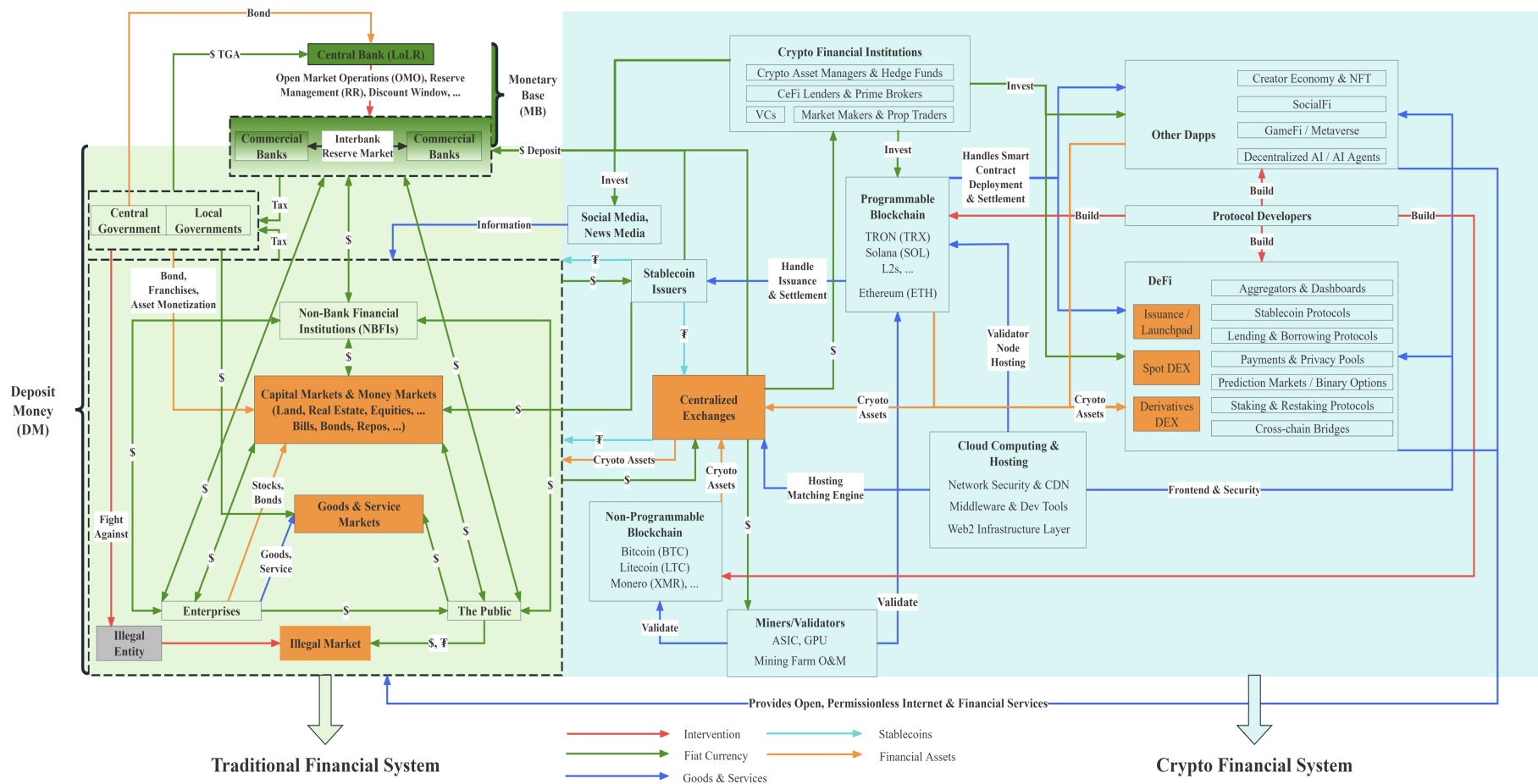


Figure 6. Structural composition of the crypto financial system and its interconnectivity with the traditional financial system

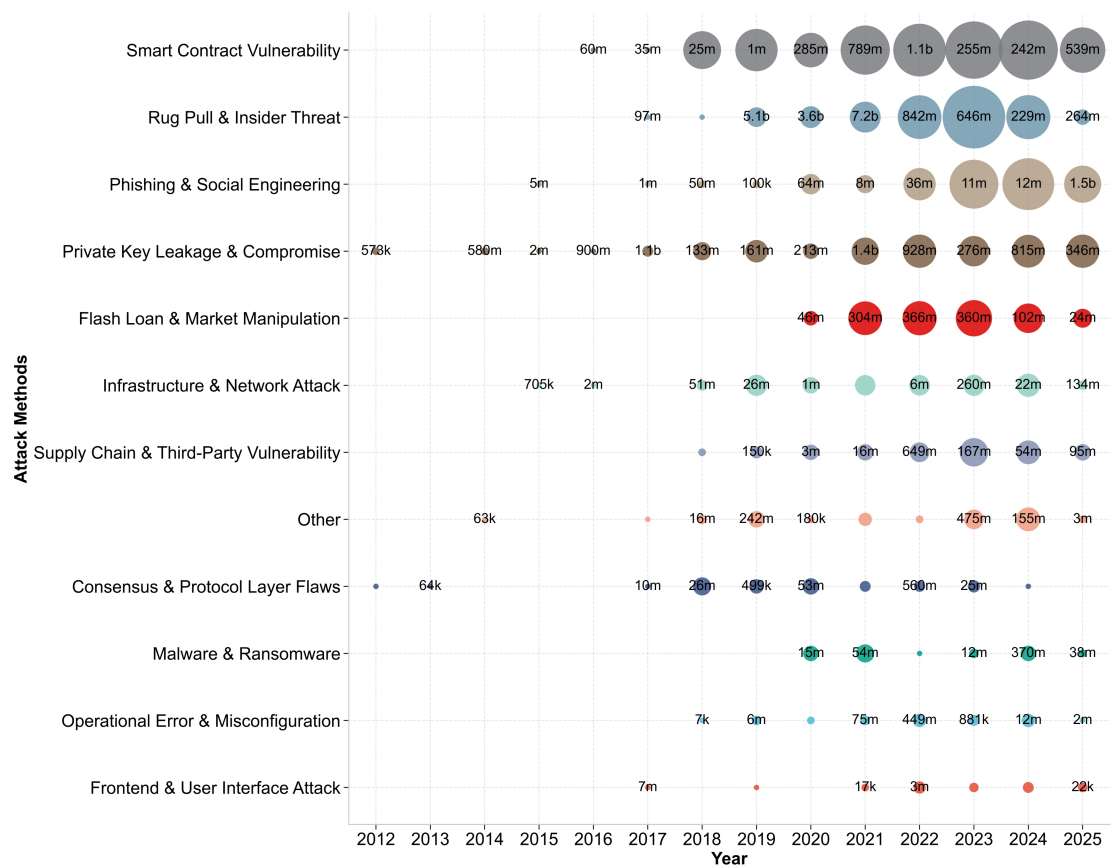


Figure 7. Crypto financial risk events aggregated by attack methods

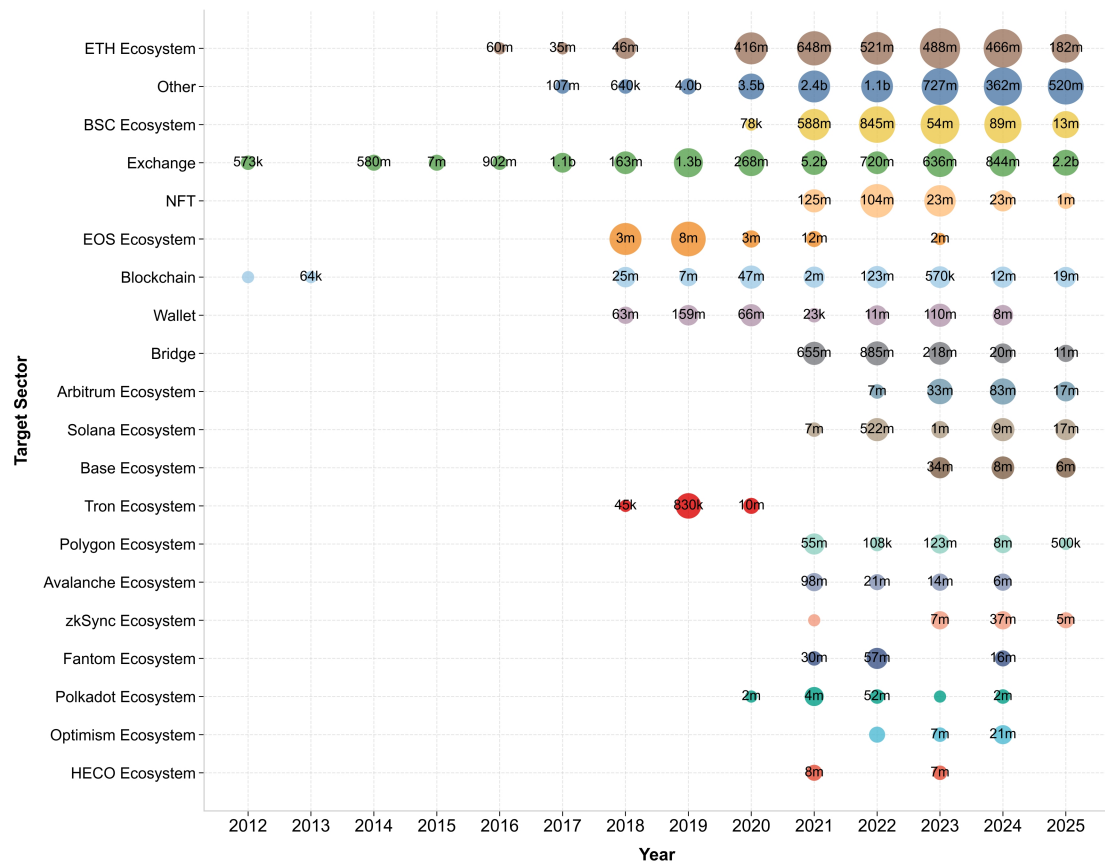


Figure 8. Crypto financial risk events aggregated by target

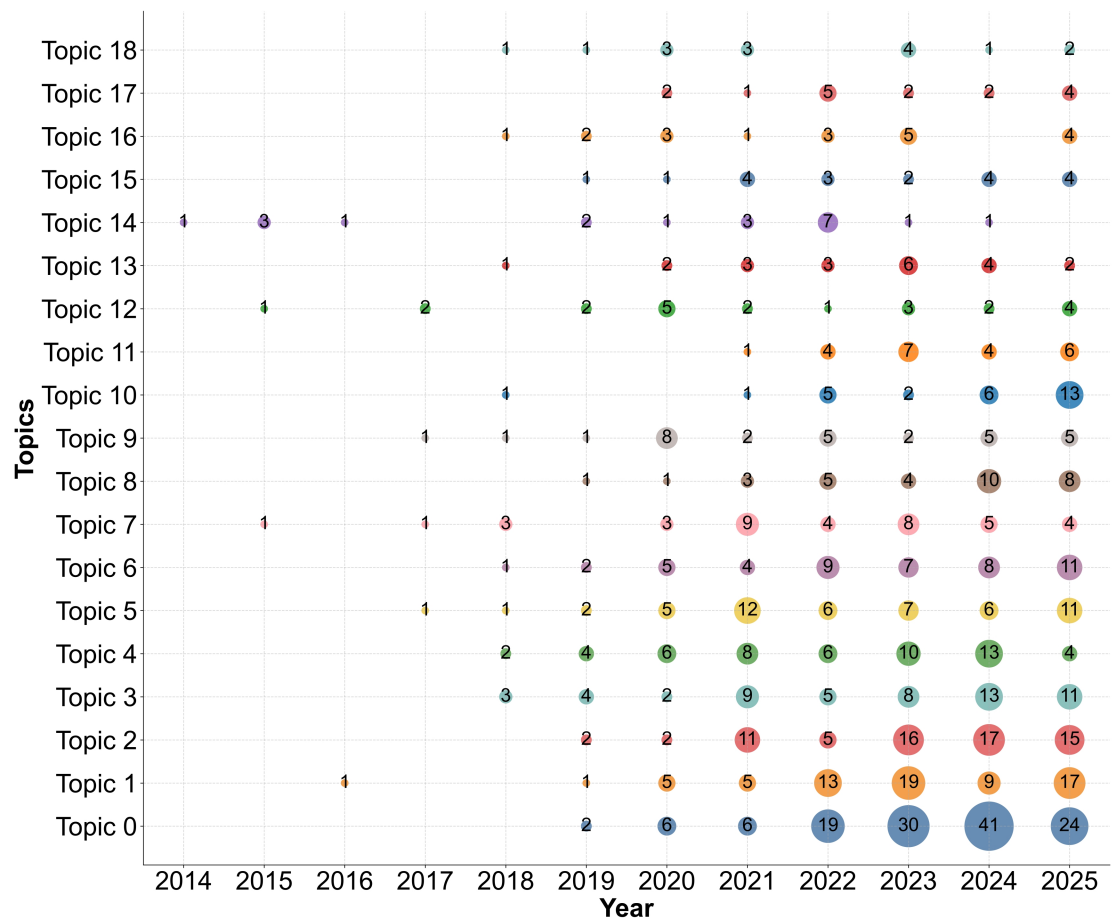


Figure 9. Results of dynamic topic analysis

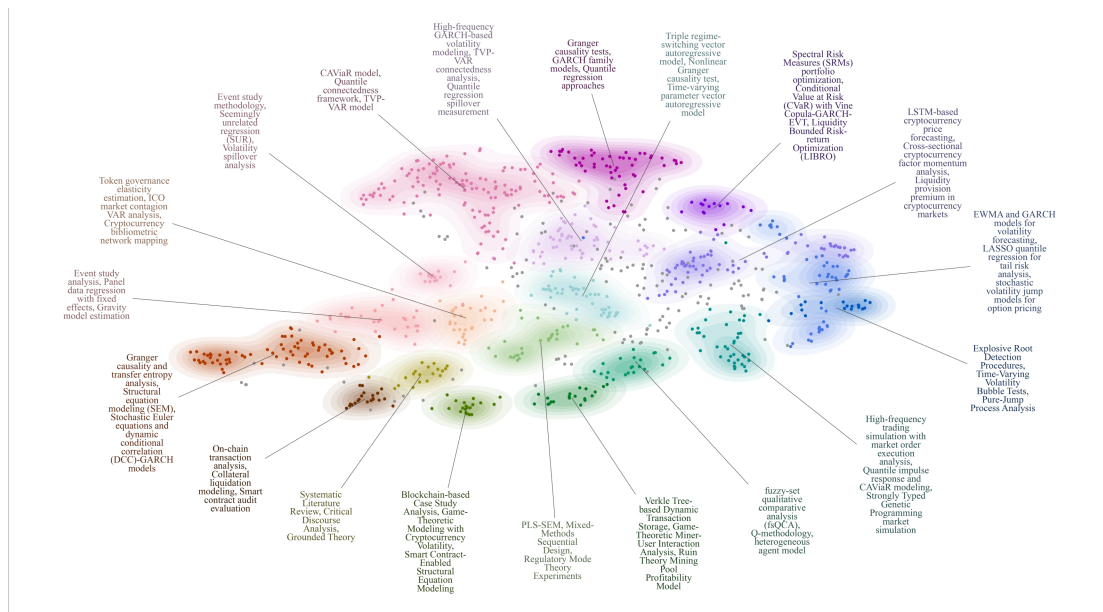


Figure 10. Clustering map of common research methodologies across topics

Appendix



Fig A.1. Crypto asset prices and major risk events in the crypto financial industry (numerically indexed)¹

¹ To ensure clarity and conciseness, only three representative events are highlighted for each year. For a full list of all incidents, please refer to Appendix Table 1.

**Table 1. Categorization framework for crypto financial risk
based on topic clustering**

Research Dimensions and the Volume of Literature	Thematic Designations	Subject Keywords	Core Focal Points
Risk Sources (453)	Topic -1: Cryptocurrency Economics and Finance	'price', 'bitcoin', 'cryptocurrencies', 'crypto', 'exchanges'	Technological vulnerabilities, institutional risks, irrational investor behaviors, and external environmental shocks
	Topic 0: Cryptocurrency and Energy Market Connectedness	'energy', 'green', 'connectedness', 'oil', 'carbon'	
	Topic 6: Investor Sentiment and Cryptocurrency Markets	'sentiment', 'investor', 'news', 'attention', 'investor attention'	
	Topic 8: Cryptocurrency User and Investor Behavior	'perceived', 'intention', 'consumers', 'ambiguity', 'adoption'	
	Topic 9: Initial Coin Offerings and Cryptocurrency Finance	'icos', 'initial coin', 'offerings', 'initial coin offerings', 'coin offerings'	
	Topic 12: Bitcoin Valuation and Price Dynamics	'exchange', 'value', 'bitcoin', 'rate', 'currency'	
	Topic 13: Blockchain Solutions for Business and Industry	'blockchain', 'smart', 'supply chain', 'supply', 'chain'	
	Topic 14: Economics of Blockchain Mining and Transaction Fees	'miners', 'fees', 'transaction', 'block', 'mining'	
Risk Measurement (70)	Topic 17: Financial Market Impact of Cryptocurrency Cyber Attacks	'attacks', 'cyber', 'estate', 'real estate', 'real'	Asset Risk Detection and Innovations in Econometric Methods
	Topic 5: Modeling Cryptocurrency Volatility and Tail Risk	'options', 'volatility', 'models', 'model', 'forecasting'	
Risk Contagion (185)	Topic 16: Econometric detection of explosive roots and financial bubbles	'explosive', 'test', 'simulations', 'bubble', 'root'	Cross-market volatility spillovers, the
	Topic 1: Crypto Assets, Financial Stability, and Central Banking	'stablecoins', 'stability', 'financial', 'bank', 'digital'	

Research Dimensions and the Volume of Literature	Thematic Designations	Subject Keywords	Core Focal Points
	Topic 3: Volatility and Spillovers in Cryptocurrency Markets	'volatility', 'connectedness', 'spillovers', 'covid19', cryptocurrencies'	accumulation of systemic risk in DeFi, and contagion to traditional finance
	Topic 7: Arbitrage and Efficiency in Bitcoin Markets	'futures', 'spot', 'arbitrage', 'bitcoin futures', 'price discovery'	
	Topic 11: DeFi Protocols and Systemic Risk	'defi', 'lending', 'finance', 'protocols', 'decentralized finance'	
Risk Hedging (136)	Topic 2: Cryptocurrency Price Prediction and Trading Strategies	'returns', 'learning', 'momentum', 'cryptocurrency', 'cryptocurrency returns'	Development of forecasting methodologies, asset valuation logic, the discernment of safe-haven properties, and portfolio risk measurement
	Topic 4: Bitcoin and Gold as Safe Haven Assets	'gold', 'safe', 'safehaven', 'hedging', 'bitcoin'	
	Topic 18: Cryptocurrency Portfolio Optimization and Risk Management	'diversification', 'portfolios', 'optimization', 'diversification benefits', 'benefits'	
Regulation & Governance (47)	Topic 10: Cryptocurrency Regulation and Illicit Finance	'flows', 'capital', 'crossborder', 'tax', 'regulation'	Anti-Money Laundering (AML)
	Topic 15: Blockchain Governance, Regulation, and Accounting	'governance', 'blockchain', 'accounting', 'regulation', 'auditing'	Regulation, On-chain Governance Structures, and Accounting and Auditing Standards

Note: Topics are assigned to primary dimensions by dominant semantic focus. Topic 1 is treated primarily under Risk Spillovers (Section 3.3) but also covers institutional-risk aspects in Section 3.1.1(2); Topic 10 is treated primarily under Regulation & Governance (Section 3.5) but also covers measurement aspects in Section 3.2; Topic 12 spans both endogenous (Section 3.1.1(1)) and exogenous (Section 3.1.1(4)) risk sources and is discussed accordingly in both sections.

Table 2. Sources of cryptographic technology and public blockchain risk

Risk Dimensions	Specific Risk Manifestations	Relevant Literature	Representative Cases
Private Key & Self-Custody	1. Insecure Generation: Low-entropy PRNGs. 2. Custodial Risk: Key loss or theft. 3. Operational Error: Erroneous address transfers.	(Erinle et al., 2025)	LuBian Pool: Brute-force of MT19937 PRNG led to 127k BTC loss.
Consensus Mechanisms	1. Loss of settlement finality: Double-spending attacks. 2. Network Congestion: Latency and execution failure. 3. Governance Risk: Contentious hard forks. 4. Block Manipulation: Strategic tampering with transaction sequencing and selection during block construction	(Capponi et al., 2025; Gans and Halaburda, 2024; Jenni et al., 2025; Jiang et al., 2022; Zhang et al., 2021)	1. BTG/ETC: Successful 51% hashpower takeovers and block reorgs (2018/2020). 2. BTC: 2017 scaling-induced forks. 3. MEV Exploitation: High-frequency arbitrage and sandwich attacks
Smart Contracts & DAO	1. Code Vulnerabilities: Flaws enabling the unauthorized exfiltration, immobilization, burning, or minting of digital assets. 2. Privilege Misuse: Compromised or exploited administrative keys allowing for unauthorized protocol interventions. 3. Governance Risk: Concentration of voting power leading to centralized manipulation of decentralized protocols.	(Sun et al., 2024)	Gala Games: Admin key breach enabled unauthorized minting of 5B GALA (2024).
Front-end & UI Layers	1. Client-side Vulnerabilities: Exploits in browser-based wallet extensions (e.g., signature spoofing). 2. Interface Exploits: Phishing and UI hijacking (e.g., script injection).	(Winter et al., 2021)	BadgerDAO: Cloudflare API hijack enabled malicious script injection; \$120M loss (2021).

Table 3. Sources of DeFi risk

Risk Dimensions	Specific Risk Manifestations	Relevant Literature	Representative Cases
Economic Models	1. Ponzi Fragility: Excessive reliance on inflationary incentives. 2. Mercenary Capital: Short-term yield farming vs. loyalty. 3. Governance-only tokens: Lack of revenue-sharing in governance tokens.	(Sockin and Xiong, 2023)	1. Uniswap (2020–2021): "Harvest-and-dump" by early LPs; UNI provided pure governance without revenue-sharing utility initially. 2. Axie Infinity (2021): Unbounded SLP emission coupled with inadequate burn/sink mechanisms led to hyperinflation.
Composability	1. Interdependency: Foundational primitive failure contagion. 2. Homogeneity: Cloned architectures share vulnerabilities.	(Saengchote, 2023)	1. Cream Finance (2021): \$130M loss via flash-loan-driven cross-protocol exploit. 2. Curve (2023): Vyper compiler bug in base pools triggered contagion in dependent protocols (Alchemix/JPEG'd).
Algorithmic Stability	1. Death Spirals: Arbitrage-driven hyperinflation and liquidity depletion.	(Briola et al., 2023)	1. Iron Finance (2021); Terra/LUNA (2022): De-pegging events triggered hyperinflationary spirals and total liquidity exhaustion.
Liquidity Provisioning	1. Impermanent Loss: Principal erosion due to price divergence in AMM pools.	(Loesch et al., 2021; Park, 2023)	1. Uniswap V3: Empirical evidence shows ~49.5% of LPs suffered net losses relative to buy-and-hold strategies due to impermanent loss.
Liquidation Mechanisms	1. Oracle Risk: Latency/stale feeds causing bad debt. 2. Incentive Failure: Gas spikes blocking liquidators. 3. Cascades: Endogenous selling pressure loops.	(Cong et al., 2025)	1. Venus Protocol (2022): Stale LUNA price feed post-collapse enabled \$13.5M bad debt exploit via oracle discrepancy.
Cross-chain Bridges	1. Centralization: Multisig or centralized governance creating single points of failure. 2. Verification: Flaws in external validators or light clients enabling proof fabrication. 3. Cross-protocol Contagion: Bridge exploits propagate shocks across ecosystems through inter-chain asset mapping and wrapped token interdependencies.	(Badev and Watsky, 2023)	1. Ronin (2022): \$610M theft via compromised validator private keys. 2. Wormhole (2022): 120k wETH unauthorized minting via signature verification bypass.

Table 4. Sources of institutional risk in CEXs

Intrinsic Characteristics	Specific Risk Manifestations	Relevant Literature
1. 24/7 Perpetual Trading	Volatility Persistence: Absence of cooling-off periods enables instantaneous sentiment manifestation. Off-peak Liquidity Risk: Market depth depletion during non-core hours increases block-trade sensitivity.	(Gemayel et al., 2023)
2. Market Fragmentation	Liquidity Fragmentation: Proliferation of exchanges and trading pairs disperses market depth, resulting in chronic illiquidity and high slippage.	(Gandal et al., 2018; Scaillet et al., 2018)
3. Regulatory Laxity	Manipulation: Prevalence of "pump-and-dump" schemes. Wash Trading: CEXs may engage in self-dealing or use matched orders to inflate trading volumes artificially. This systematic fabrication of activity misleads investors regarding genuine liquidity depth and asset demand.	(Aloosh and Li, 2024; Cong et al., 2023b; Li et al., 2025)
4. Excessive Leverage	Liquidity-Leverage Resonance: Systematic liquidation of high-leverage positions under thin market depth, triggering self-reinforcing slippage and cascading "liquidation spirals."	(Alexander et al., 2023)
5. Centralized Custody	Misappropriation: Commingling or clandestine use of client funds for speculative proprietary ventures. Operational and Cybersecurity: Massive asset concentration attracts targeted cyberattacks against centralized infrastructure.	(Aquilina et al., 2023; Vidal-Tomás, 2025)
6. Listing Authority	Informational Asymmetry and Moral Hazard: Leveraging absolute discretionary listing authority, CEXs may admit substandard assets to the market while colluding with market makers to manipulate prices.	(Financial Stability Board, 2023)
7. Primary Price Feeds	Oracle Contagion: Since many DeFi protocols rely on CEX data as oracle sources, manipulated or anomalous exchange prices may trigger cascading liquidations and systemic losses within the DeFi ecosystem.	(Duley et al., 2023; Werner et al., 2022)

Table 5: Sources of stablecoin issuer risk

Risk Dimensions	Specific Risk Manifestations	Relevant Literature
1. Credit & Opacity	1. Historical Asset Substitution: During early market phases, issuers may replace high-quality liquid assets with speculative-grade paper or opaque inter-affiliate loans. 2. Audit Latency: Information asymmetry persists due to the lag in third-party attestations, masking real-time fair value and reserve quality.	(Galati and Capalbo, 2024; Maex and Slavov, 2025)
2. Liquidity Risk	1. Gateway Fragility: High banking partner concentration creates redemption bottlenecks; bank-level shocks can suspend fiat-to-crypto flows, converting liquidity crunches into terminal de-pegging events. 2. Maturity Mismatch: Liquidity transformation via long-duration instruments impairs instantaneous redemption capacity.	(Gregory et al., 2024; Ma et al., 2023)
3. Market & De-pegging	1. Utility Deficit & Market Runs: Lack of organic use-cases renders non-dominant stablecoins dependent on speculative demand; secondary-market illiquidity during systemic panic transforms transient volatility into terminal de-pegging spirals. 2. Collateral Impairment: Macroeconomic shifts (e.g., rate hikes or recessionary contractions) induce mark-to-market devaluation of reserves, compromising solvency.	(Gorton et al., 2025)
4. Systemic Risk	1. Network Criticality: Systemic importance to global crypto asset trading, DeFi protocols, and cross-border payment systems. 2. LOLR Absence: Lack of a Lender of Last Resort to provide emergency liquidity. 3. Cross-border Velocity: High-frictionless mobility facilitates rapid capital flight and monetary instability. 4. Financial Linkages: Strong interconnectedness with money markets.	(Ahmed and Aldasoro, 2025; Azzimonti and Quadrini, 2025)

Table 6: Sources of crypto asset investment institution risk

Risk Dimensions	Specific Risk Manifestations	Representative Cases
1. Leverage & Liquidity	1. Deleveraging Spirals: High-leverage liquidations during market reversals. 2. Maturity Transformation: Funding illiquid long-term assets with short-term deposits, creating run vulnerability.	1. Three Arrows Capital(2022,6): High-leverage strategies were subject to cascading liquidations. 2. Celsius Network (2022,7): Liquidity exhaustion via stETH de-pegging and withdrawal freezes.
2. Credit & Counterparty	1. Domino Effects: Rapid propagation of insolvency through inter-institutional credit-debt linkages across critical nodes. 2. Under-collateralization: Extensive unsecured lending between major nodes.	1. Voyager Digital(2022,7): Insolvent via \$670M default by Three Arrows Capital. 2. Genesis Trading (2023,1): \$1B+ bad debt from Three Arrows Capital and Alameda collapses.
3. Concentration & Manipulation	1. Concentration Risk: Excessive portfolio sensitivity to a narrow subset of assets or counterparty loan exposures. 2. Strategic Manipulation: Collusive practices and fraudulent trading to distort market integrity and pricing.	1. Alameda Research (2022,11): Insolvency via concentrated exposure to illiquid affiliated tokens (FTT). 2. Jump Trading(2021-2022): Alleged market manipulation to support UST peg prior to total collapse.
4. Governance & Opacity	1. Governance Failure: Excessive centralization and lack of internal risk controls. 2. Information Opacity: Inadequate disclosure and risk-exposure concealment.	1. FTX (2022,11): Total collapse of corporate controls; unauthorized diversion of client funds for proprietary trading. 2. Three Arrows Capital(2022): Deliberate falsification of balance sheets to conceal extreme leverage and loss positions.
5. Systemic Contagion	1. Liquidity Spiral: Panic-induced feedback loops where mass redemptions and forced liquidations drive self-reinforcing price collapses (Esparcia et al., 2024).	1. SVB Failure (2023): Traditional banking failure triggered a reserve impairment, causing a transient USDC de-pegging.

Table 7. Sources of exogenous risks

Risk Dimensions	Risk Manifestations	Representative Cases
1. Cloud Infrastructure	1. Operational Fragility: DApp/CEX reliance on centralized RPC nodes; cloud outages disrupt transaction execution and deposits. 2. Network Degradation: Cloud-dependent sequencers/validators cause TPS drops and liquidation failures during outages.	AWS (2025): Coinbase and Base network disruptions due to sequencer and node dependency on AWS.
2. Internet Resources	1. Access Disruption: Domain/API compromise blocks user interaction, triggering panic. 2. Off-chain data contamination: DNS attacks against blockchain explorers may deliver false on-chain data and mislead investors.	Infura Outage (2022): RPC failure disabled MetaMask and major DApp data queries and transactions.
3. Regulatory Policy	1. Legal Uncertainty: Sudden enforcement actions against issuers/exchanges induce price shocks. (Auer and Claessens, 2018); 2. Operational Viability: Evolving compliance mandates redefine the cost and legitimacy of services, potentially obsoleting existing business models.	Tornado Cash (2025): The U.S. Treasury's removal of sanctions on Tornado Cash drove price and volume increases across privacy-focused cryptocurrencies.
4. Monetary Policy	1. Valuation Compression: Rate hikes reduce risk appetite and market liquidity (Che, 2023). 2. Macro-financial Linkages: Tightening increases financing costs, exacerbating institutional maturity mismatches.	Fed Tightening (2022): Global rate hikes triggered systematic deleveraging and liquidity crises.
5. Geopolitical Conflict	1. Geopolitical Shocks: Military conflicts often trigger crypto liquidation and market declines. 2. Sanctions Enforcement: Geopolitical mandates leading to address freezing, blacklisting, or cross-border service suspension.	Russia-Ukraine (2022): Major CEXs restricted Russian accounts post-sanctions. Israel-Palestine (2024): Coordinated freezing of Hamas-linked crypto addresses.
6. Energy Policy	1. Security Shocks: Mining bans cause abrupt hashrate declines, weakening network resilience and increasing short-term volatility.	China Ban (2021): Bitcoin hashrate collapsed by ~50%, exerting acute downward market pressure.

Table A.1. Representative Large-scale Loss Events in the Crypto Financial

Industry			
No.	Date	Event Name	Loss Amount
1	2017-04-27	Yapizon Exchange Security Breach	3831 BTC
2	2017-07-19	Parity Wallet Vulnerabilities	~150,000 ETH
3	2017-11-19	The Tether (USDT) Treasury Security Breach	~\$31 Million (USDT)
4	2018-01-26	The Coincheck Exchange Security Breach	~\$530 Million
5	2018-02-12	The BitGrail Exchange Security Breach and Insolvency	~\$195 Million
6	2018-07-10	The Bancor DEX Security Breach	~\$13.5 Million
7	2019-05-07	The 2019 Binance Security Breach	7074 BTC
8	2019-06-29	The PlusToken Ponzi Scheme	~\$3 Billion
9	2019-11-27	The Upbit Exchange Security Breach	~342,000 ETH
10	2020-02-17	The FCoin Exchange Insolvency and Liquidity Collapse	Total Insolvency Gap of 7,000-13,000 BTC
11	2020-08-30	Electrum Wallet Phishing Attack	~1400 BTC
12	2020-09-25	The KuCoin Exchange Security Breach	~\$150 Million
13	2021-08-10	The Poly Network Cross-chain Protocol Exploit	~\$611 Million
14	2021-10-27	Cream Finance Flash Loan Exploits	~\$130 Million
15	2021-12-04	The BitMart Exchange Security Breach	~\$150 Million
16	2022-03-29	The Ronin Network Sidechain Security Breach	~\$624 Million
17	2022-05-08	The Collapse of the Terra/Luna Ecosystem	~\$40 Billion
18	2022-11-09	The Collapse and Bankruptcy of the FTX Exchange	~\$8 Billion
19	2023-03-13	The Euler Finance Protocol Exploit	~\$197 Million
20	2023-07-30	The Curve Finance Vyper Compiler Exploit	~\$70 Million
21	2023-09-23	The Mixin Network Cloud Infrastructure Breach	~\$200 Million
22	2024-03-28	The Prisma Finance Protocol Exploit	~\$11.5 Million
23	2024-05-20	The Gala Games Minting Exploit	~\$21.8 Million
24	2024-05-31	The 2024 DMM Bitcoin Security Breach	~\$305 Million
25	2025-02-21	The 2025 Bybit Cold Wallet Supply Chain Attack	~\$1.4 Billion
26	2025-05-22	The 2025 Cetus Protocol Exploit	~\$223 Million
27	2025-11-03	Balancer V2 Logic Vulnerability	~\$121 Million

Table A.2. Parameter selection for the model

Key Component	Setting / Model
Dimensionality Reduction (UMAP)	n_neighbors: int = 15, n_components: int = 2, metric: str = "euclidean"
Clustering (HDBSCAN)	min_cluster_size=15
Vectorizer Model	ngram_range=(1, 3), stop_words=english_stop_words
CTFIDF Model	bm25_weighting=True, reduce_frequent_words=True
Topic Representation	gpt-5
	gemini-2.5-pro
	deepseek-v3.2

Table A.3. Topic representation based on Text-Embedding-3-Large

Topic	Count	MMR	POS	GPT	Gemini	Deepseek
-1	137	['price', 'bitcoin', 'cryptocurrencies', 'crypto', 'exchanges']	['price', 'bitcoin', 'cryptocurrencies', 'crypto', 'exchanges']	['Cryptocurrency Markets, Pricing Dynamics, and Investment Risks']	['Cryptocurrency Economics and Finance']	['Cryptocurrency Market Dynamics and Economic Implications']
0	128	['energy', 'green', 'connectedness', 'oil', 'carbon']	['energy', 'green', 'connectedness', 'oil', 'carbon']	['Energy and Green Asset Connectedness in Cryptocurrency and Commodity Markets']	['Cryptocurrency and Energy Market Connectedness']	['Green Cryptocurrency and Energy Market Connectedness in Financial Risk and Sustainability']
1	70	['stablecoins', 'stability', 'financial', 'bank', 'digital']	['stablecoins', 'stability', 'financial', 'bank', 'digital']	['Central Bank Digital Currencies, Stablecoins, and Cryptoasset Risks in Financial Systems']	['Crypto Assets, Financial Stability, and Central Banking']	['Financial Stability and Digital Currency Risks in Emerging Markets']
2	68	['returns', 'learning', 'momentum', 'cryptocurrency', 'cryptocurrency returns']	['returns', 'learning', 'cryptocurrency', 'momentum', 'machine']	['Machine learning and quantitative strategies in cryptocurrency markets']	['Cryptocurrency Price Prediction and Trading Strategies']	['Cryptocurrency Market Analysis and Predictive Modeling']
3	55	['volatility', 'connectedness', 'spillovers', 'covid19', 'cryptocurrencies']	['volatility', 'connectedness', 'spillovers', 'conditional', 'cryptocurrencies']	['Cryptocurrency volatility and spillover dynamics across markets and networks during crises']	['Volatility and Spillovers in Cryptocurrency Markets']	['Cryptocurrency volatility spillovers and market connectedness']
4	53	['gold', 'safe', 'safehaven', 'hedging', 'bitcoin']	['gold', 'safe', 'safehaven', 'hedging', 'bitcoin']	['Bitcoin and Gold as Safe Haven and Hedging Assets in Global Markets']	['Bitcoin and Gold as Safe Haven Assets']	['Cryptocurrency and Gold as Safe Haven Assets and Hedging Instruments']
5	51	['options', 'volatility', 'models', 'model', 'forecasting']	['options', 'volatility', 'models', 'model', 'forecasting']	['Cryptocurrency volatility modeling and risk measurement techniques']	['Modeling Cryptocurrency Volatility and Tail Risk']	['Financial Risk Modeling and Volatility Forecasting in Cryptocurrency Markets']
6	47	['sentiment', 'investor', 'news', 'attention', 'investor attention']	['sentiment', 'investor', 'news', 'attention', 'media']	['Impact of investor sentiment and media on cryptocurrency markets and volatility']	['Investor Sentiment and Cryptocurrency Markets']	['Sentiment and Investor Behavior in Cryptocurrency Markets']
7	38	['futures', 'spot', 'arbitrage', 'bitcoin']	['futures', 'spot', 'arbitrage', 'discovery', 'introduction']	['Bitcoin market structure, efficiency, and arbitrage in spot and	['Arbitrage and Efficiency in Bitcoin	['Bitcoin Market Microstructure and Arbitrage

Topic	Count	MMR	POS	GPT	Gemini	Deepseek
		futures', 'price discovery']		futures markets']	Markets']	Dynamics']
8	32	['perceived', 'intention', 'consumers', 'ambiguity', 'adoption']	['intention', 'consumers', 'ambiguity', 'adoption', 'cryptocurrency']	['Consumer behavior, psychological factors, and risk perceptions in cryptocurrency adoption and investment']	['Cryptocurrency User and Investor Behavior']	['Cryptocurrency Adoption and Investment Behavior']
9	30	['icos', 'initial coin', 'offerings', 'initial coin offerings', 'coin offerings']	['initial coin', 'offerings', 'initial', 'research', 'coin']	['Cryptocurrency and ICO Governance, Market Dynamics, and Disruptive Financial Innovations']	['Initial Coin Offerings and Cryptocurrency Finance']	['Cryptocurrency and Fintech Innovation in Alternative Finance']
10	28	['flows', 'capital', 'crossborder', 'tax', 'regulation']	['flows', 'capital', 'tax', 'regulation', 'regulatory']	['Global Regulation, Illicit Use, and Economic Impacts of Cryptocurrencies']	['Cryptocurrency Regulation and Illicit Finance']	['Cryptocurrency Regulation and Cross-Border Capital Flows']
11	22	['defi', 'lending', 'finance', 'protocols', 'decentralized finance']	['lending', 'finance', 'protocols', 'decentralized', 'decentralized finance']	['Decentralized Finance Systems, Risks, and Applications']	['DeFi Protocols and Systemic Risk']	['Decentralized Finance Systems and Challenges']
12	22	['exchange', 'value', 'bitcoin', 'rate', 'currency']	['exchange', 'value', 'bitcoin', 'rate', 'currency']	['Bitcoin valuation, price dynamics, and market behavior']	['Bitcoin Valuation and Price Dynamics']	['Bitcoin Price Dynamics and Value Systems']
13	21	['blockchain', 'smart', 'supply chain', 'supply', 'chain']	['blockchain', 'smart', 'supply', 'chain', 'smart contract']	['Blockchain Applications and Smart Contract Solutions for Supply Chain, Finance, Healthcare, and Infrastructure Management']	['Blockchain Solutions for Business and Industry']	['Blockchain Applications in Industry and Supply Chain Management']
14	20	['miners', 'fees', 'transaction', 'block', 'mining']	['miners', 'fees', 'transaction', 'block', 'mining']	['Bitcoin mining economics, incentives, and transaction fee dynamics']	['Economics of Blockchain Mining and Transaction Fees']	['Bitcoin Mining Economics and Transaction Fee Dynamics']
15	19	['governance', 'blockchain', 'accounting', 'regulation', 'auditing']	['governance', 'blockchain', 'accounting', 'regulation', 'auditing']	['Blockchain governance, trust, and regulation in financial and accounting systems']	['Blockchain Governance, Regulation, and Accounting']	['Blockchain Governance and Trust in Financial Systems']
16	19	['explosive', 'test', 'simulations', 'bubble', 'root']	['explosive', 'test', 'simulations', 'bubble', 'root']	['Detection and Modeling of Explosive Behavior and Bubbles in Financial Time Series']	['Econometric detection of explosive roots and financial bubbles']	['Econometric Methods for Financial Bubble Detection and Volatility Modeling']

Topic	Count	MMR	POS	GPT	Gemini	Deepseek
17	16	['attacks', 'cyber', 'estate', 'real estate', 'real']	['attacks', 'cyber', 'real estate', 'estate', 'real']	['Impacts of Cyberattacks and Cybercrime on Financial Markets and Assets']	['Financial Market Impact of Cryptocurrency Cyber Attacks']	['Cybersecurity and Privacy Breaches Impact on Financial Markets and Digital Assets']
18	15	['diversification', 'portfolios', 'optimization', 'diversification benefits', 'benefits']	['diversification', 'portfolios', 'benefits', 'optimization', 'estimation']	['Cryptocurrency portfolio diversification and optimization strategies']	['Cryptocurrency Portfolio Optimization and Risk Management']	['Cryptocurrency Portfolio Diversification and Risk Management']

Table A.4. Topic representation based on Gemini-Embedding-001

Topic	Count	MMR	POS	GPT	Gemini	Deepseek
-1	124	['bitcoin', 'bubble', 'explosive', 'price', 'bubbles']	['bitcoin', 'bubble', 'explosive', 'price', 'bubbles']	['Cryptocurrency Markets, Price Bubbles, and Financial Risk Analysis']	['Cryptocurrency Price Bubbles and Market Volatility']	['Cryptocurrency Market Dynamics and Risk Analysis']
0	88	['green', 'energy', 'connectedness', 'carbon', 'clean']	['green', 'energy', 'connectedness', 'carbon', 'clean']	['Green and Energy-Related Financial Market Connectedness, Spillovers, and Hedging with Cryptocurrencies and Sustainable Assets']	['Connectedness Between Cryptocurrencies and Green Energy Markets']	['Green Energy and Cryptocurrency Market Connectedness']
1	80	['gold', 'oil', 'safe', 'diversification', 'bitcoin']	['gold', 'oil', 'safe', 'diversification', 'bitcoin']	['Diversification, risk management, and safe-haven roles of gold, oil, and cryptocurrencies in portfolio investment and financial markets']	['Cryptocurrencies and Commodities as Safe Haven and Hedging Assets']	['Financial Commodities and Cryptocurrency Diversification Strategies']
2	60	['spillovers', 'cryptocurrencies', 'connectedness', 'conditional', 'spillover']	['spillovers', 'cryptocurrencies', 'connectedness', 'conditional', 'spillover']	['Volatility spillovers, risk connectedness, and contagion dynamics in cryptocurrency and global financial markets']	['Volatility Spillovers and Connectedness in Cryptocurrency Markets']	['Cryptocurrency Volatility Spillovers and Network Risk Dynamics']
3	54	['futures', 'liquidity', 'spot', 'exchanges', 'bitcoin futures']	['futures', 'liquidity', 'spot', 'exchanges', 'arbitrage']	['Liquidity, Trading Dynamics, and Price Discovery in Cryptocurrency Spot and Futures Markets']	['Liquidity and Price Discovery in Bitcoin Spot and Futures Markets']	['Cryptocurrency Market Microstructure and Trading Dynamics']
4	53	['sentiment', 'investor', 'news', 'attention', 'investor attention']	['sentiment', 'investor', 'news', 'attention', 'herding']	['Behavioral factors, sentiment, and investor attention in cryptocurrency markets']	['Investor Sentiment and Attention in Cryptocurrency Markets']	['Investor Psychology and Behavioral Dynamics in Cryptocurrency Markets']
5	51	['crypto', 'crossborder', 'crypto assets', 'financial', 'stablecoins']	['crypto', 'crossborder', 'crypto assets', 'financial', 'stablecoins']	['Digital Assets and Financial Stability: Risks, Cross-Border Flows, Stablecoins, and Central Bank Policy']	['Digital Currencies, Cross-Border Payments, and Financial Stability']	['Cryptoassets and Digital Finance: Financial Stability, Cross-Border Flows, and Policy Challenges']
6	47	['options', 'volatility', 'models', 'model', 'forecasting']	['options', 'volatility', 'models', 'model', 'forecasting']	['Volatility modeling and option pricing in cryptocurrency markets']	['Cryptocurrency Volatility Forecasting and Option Pricing']	['Cryptocurrency Volatility Modeling and Forecasting']
7	43	['mining', 'miners', 'transaction', 'fees']	['mining', 'miners', 'transaction', 'fees']	['Bitcoin and Cryptocurrency Economics, Transaction Dynamics,']	['Cryptocurrency and Blockchain Economics']	['Bitcoin Economics and Scalability']

Topic	Count	MMR	POS	GPT	Gemini	Deepseek
		'block']	'block']	and Protocol Design']		
8	36	['intention', 'perceived', 'adoption', 'consumers', 'consumer']	['intention', 'adoption', 'consumers', 'consumer', 'ambiguity']	['Consumer Behavior, Risks, and Social Influences in Cryptocurrency Adoption and Usage']	['Consumer Behavior and Social Influence in Cryptocurrency Adoption']	['Cryptocurrency Adoption and User Behavior']
9	35	['learning', 'machine', 'machine learning', 'forecasting', 'determinants']	['learning', 'machine', 'forecasting', 'determinants', 'models']	['Machine Learning and Deep Learning Approaches for Cryptocurrency Price Prediction and Trading Strategies']	['Machine Learning Models for Cryptocurrency Forecasting']	['Cryptocurrency Price Forecasting and Trading Strategies Using Machine Learning']
10	30	['momentum', 'returns', 'cryptocurrency', 'cryptocurrency returns', 'factor']	['momentum', 'returns', 'cryptocurrency', 'factor', 'risk']	['Cross-Sectional Risk Factors, Momentum, and Performance Measurement in Cryptocurrency Returns and Funds']	['Cryptocurrency Returns, Risk Factors, and Fund Performance']	['Cryptocurrency Risk Factors and Return Predictability']
11	27	['defi', 'lending', 'protocols', 'finance defi', 'finance']	['lending', 'protocols', 'finance', 'decentralized', 'decentralised']	['Decentralized Finance (DeFi) Systems, Lending Protocols, and Market Risks']	['Decentralized Finance and Lending Protocols']	['Decentralized Finance (DeFi) Systems and Challenges']
12	26	['multifractal', 'volatility', 'asymmetric', 'efficiency', 'market efficiency']	['multifractal', 'volatility', 'asymmetric', 'efficiency', 'structural breaks']	['Multifractal and Asymmetric Volatility Dynamics in Cryptocurrency Markets']	['Multifractality, Volatility, and Efficiency in Cryptocurrency Markets']	['Cryptocurrency market dynamics and volatility modeling']
13	25	['stablecoins', 'reserve', 'stability', 'stable', 'tether']	['stablecoins', 'reserve', 'stability', 'stable', 'dollar']	['Stablecoin Market Dynamics, Risks, and Peg Stability']	['Stablecoin Peg Stability, Runs, and Devaluation Risks']	['Stablecoin Stability and Market Dynamics']
14	23	['blockchain', 'supply', 'supply chain', 'chain', 'smart']	['blockchain', 'supply', 'chain', 'smart', 'technology']	['Blockchain Technology Applications in Supply Chains, Digital Asset Management, and Transparency Solutions']	['Blockchain Technology in Supply Chain and Information Management']	['Blockchain Applications in Supply Chains and Digital Systems']
15	21	['icos', 'offerings', 'initial coin', 'initial coin offerings', 'coin offerings']	['offerings', 'initial coin', 'initial', 'coin', 'entrepreneurs']	['Governance, Regulation, and Market Dynamics of Initial Coin Offerings and Tokenized Assets']	['Regulation, Governance, and Fraud in Initial Coin Offerings and Cryptocurrencies']	['Cryptocurrency Regulation and Initial Coin Offerings']
16	20	['research', 'tax', 'fintech', 'literature',	['research', 'tax', 'fintech', 'literature',	['Cryptocurrency, Fintech, and Alternative Finance: Research Trends,	['Research on Cryptocurrency, FinTech,	['Emerging Trends in Cryptocurrency and Fintech

Topic	Count	MMR	POS	GPT	Gemini	Deepseek
		'alternative finance']	'alternative finance']	Disruptive Impacts, and Taxation Issues']	and Taxation']	Research']
17	17	['accounting', 'regulation', 'blockchain', 'auditing', 'government']	['accounting', 'regulation', 'blockchain', 'auditing', 'government']	['Blockchain and Cryptocurrencies: Governance, Regulation, Value Systems, and Accounting Practices']	['Blockchain in Accounting, Auditing, and Governance']	['Blockchain in Accounting and Financial Systems']
18	16	['defi', 'nfts', 'nft', 'defi assets', 'cbdc']	['metaverse', 'assets', 'index', 'connectedness', 'tokens']	['Spillovers and Connectedness among DeFi, NFTs, CBDCs, and Traditional Financial Markets']	['DeFi, NFT, and CBDC Market Dynamics']	['Digital Asset Markets: DeFi, NFTs, and CBDCs Interconnectedness and Spillovers']
19	15	['uncertainty', 'geopolitical', 'economic policy', 'economic', 'gpr']	['uncertainty', 'geopolitical', 'economic policy', 'economic', 'global']	['Cryptocurrency as a Safe Haven and Hedge under Economic Policy and Geopolitical Uncertainty']	['Impact of Geopolitical Risk and Economic Policy Uncertainty on Bitcoin']	['Bitcoin and Cryptocurrency Responses to Geopolitical and Economic Uncertainty']

Table A.5. Channels for systemic risk transmission in crypto finance

Channel	Transmission Mechanisms
1. Asset Linkage	1. Correlation Surge: Stress periods see a sharp rise in correlations between major crypto assets (e.g., Bitcoin) and traditional equities (notably tech stocks), amplifying cross-market risk resonance. 2. Collateral-Liquidation Spiral: Falling crypto asset prices trigger collateral liquidations in lending markets, leading to forced sales that further depress prices in a self-reinforcing spiral. 3. Stablecoin De-pegging Impact: As key quasi-fiat units, stablecoin de-pegging erodes DeFi liquidity foundations and precipitates widespread market panic.
2. Institutional Exposure	1. Crypto-Finance Services by Traditional Institutions: Banks, hedge funds, and insurers gain exposure through direct investment, market-making, custody, or derivative services. Resulting losses can erode the capital base of the traditional financial system (Auer R. et al., 2022). 2. Interconnected Crypto Entity Contagion: The collapse of crypto hedge funds or lending platforms transmits risk through inter-entity lending networks to core ecosystem participants, potentially affecting traditional institutions via redemption pressures. 3. Crypto Treasury Company (DAT) Risk: Public companies holding significant crypto assets face balance sheet volatility, transmitting crypto market risk to equity markets through stock price declines and potential asset sales.
3. Banking System	1. Impact on the Banking System: Stablecoin issuance can compress net interest margins and lead to bank balance sheet contraction (Chen and Phelan, 2025; Liao and Caramichael, 2022).
4. Regulatory Spillover	Cross-border Regulatory Spillover & Arbitrage: Regulatory actions in major economies produce significant cross-border spillovers, shaping global capital flows and market behavior. Divergent standards across jurisdictions incentivize regulatory arbitrage, concentrating risks in under-regulated areas and ultimately threatening global financial stability (Kokenyne et al., 2022; Marcelino et al., 2023).
5. Information Expectation	1. Social Media Amplification Effect: The crypto market's heavy reliance on social media for information dissemination allows misinformation or panic to spread rapidly, amplifying market volatility. 2. Self-fulfilling Risk Expectations: Widespread expectations of specific risk events (e.g., CEX runs, stablecoin de-peg) can prompt coordinated investor actions, materializing the anticipated risks.
6. Emerging Economies	Exacerbates currency substitution effects and capital outflows while simultaneously weakening monetary policy transmission mechanisms (Copestake, 2023).